Application of Scattering Transform on simulated and actual images of galaxies to ascertain properties of galaxy mergers

Rwitobaan Sheikh

Master of Science in Machine Learning and Autonomous Systems
The University of Bath
2023/2024

Application of Scattering Transform on simulated and actual images of galaxies to ascertain properties of galaxy mergers.

Submitted by: Rwitobaan Sheikh

Copyright

Declaration

This dissertation is submitted to the University of Bath in accordance with the requirements of the degree of Master of Science in the Department of Computer Science. No portion of the work in this dissertation has been submitted in support of an application for any other degree or qualification of this or any other university or institution of learning. Except where specifically acknowledged, it is the work of the author.

Abstract

This study investigates the application of scattering transforms to both simulated and real images of galaxy mergers, aiming to enhance the classification and understanding of such mergers. Galaxy mergers, significant events in cosmic evolution, present classification challenges due to complex morphologies, varied stages, and observational distortions. Scattering transforms offer a robust analytical tool due to their translation invariance, multiscale analysis capabilities, and effective feature extraction. The project involves developing and implementing a scattering transform network on simulated data from the Illustris TNG project, which provides comprehensive simulated galaxy images and compares the results with real images from the SDSS dataset (Sloan Digital Sky Survey). The methodology includes leveraging the Kymatio library for scattering transforms, which supports various wavelet forms and integrates with deep learning architectures. This research aims to bridge the gap between simulated and real data by identifying discriminative features that can accurately classify and elucidate the properties of galaxy mergers, potentially contributing to the broader field of galaxy evolution studies.

Contents

List of Figures

List of Tables

Acknowledgements

Acknowledgements given to my supervisors Dr. Georgios Exarchakis and Dr. Carolin Villforth for their continued support throughout the project. Special thanks also given to Mathilda Avirett-Mackenzie for her help on selecting the dataset for real (sdss galaxy images) and simulated (illustris-tng galaxy images) data and implementation of Sky subtraction in the data in this project.

Chapter 1

Introduction

1.1 Comparisons of properties of real and simulated data

This study presents a comparative analysis of the structural and morphological properties of a real galaxy image and a set of simulated galaxy models. The primary objective is to infer the merger characteristics of the observed galaxy and classify its current merger status. To achieve this, we utilize the scattering transform filter, an advanced technique capable of capturing multi-scale geometric features in astronomical images. By applying the scattering transform to both real and simulated galaxy images, we extract edge-based features that effectively characterize the complex structures typically associated with galactic mergers [3].

The similarity between the real and simulated galaxies is quantitatively assessed by computing the distance between the edge features produced by the scattering transform. This metric allows us to identify the five simulated galaxies that most closely resemble the real galaxy in terms of structural properties [5]. The results from this comparison provide key insights into the dynamic processes governing the observed galaxy, enabling us to infer its merger history and current evolutionary stage. This methodology offers a novel approach for systematically distinguishing merger and non-merger features in galaxies and identifying structurally similar galaxies based on these features.

Our approach draws upon the methodology outlined in [6], originally developed for predicting molecular properties, and applies it to galaxy images. This was chosen over traditional machine learning and deep learning algorithms, which typically require high-resolution images and rely heavily on the visual quality of data. In contrast, real galaxy images, especially those of high-redshift galaxies, are often low-resolution due to the vast distances involved, spanning thousands to millions of light-years. Obtaining high-resolution images of such distant galaxies remains a significant challenge.

While deep learning has been employed in previous astronomical data comparisons and classifications, these models often suffer from incorrect predictions [7]. The scattering transform, however, is not constrained by image resolution and can derive meaningful predictions based on edge features in the images [3]. This approach aligns with the need to recognise the sparse structural features of distant galaxies.

1.2 Galaxy Mergers and problems with classification

The classification of galaxy mergers presents an immensely formidable challenge due to a multitude of contributing factors that compound the inherent difficulties involved [7]. Among the primary reasons for these challenges are the intricate and highly varied morphologies exhibited throughout the entire merging process, which often result in significant distortions of the galaxies' original shapes. These distortions frequently lead to highly irregular and complex appearances that deviate substantially from typical galactic forms [1]. The different stages of galaxy mergers, ranging from initial interaction to final coalescence, further exacerbate the classification difficulty. Each stage can manifest in distinctly different observable features, which vary depending on the specific conditions of the merger. Furthermore, the varying mass ratios between the merging galaxies introduce yet another layer of complexity [8]. These varying mass ratios create a diverse array of observable phenomena, which can further obscure the accurate categorization of galaxy mergers. The situation is further complicated by projection effects, which are the result of our perspective when observing the mergers. These effects can significantly alter the apparent structure of the galaxies and their interactions, often leading to interpretations that may not accurately reflect the true nature of the merging process. Additionally, interactions with the surrounding environment, such as gravitational influences from nearby structures or intergalactic material, can induce further variations in the observable features. These environmental interactions can produce structures that closely mimic those typically associated with merger signatures, thereby adding another layer of difficulty to the classification process [8]. Redshift effects, which refer to the shift in the wavelength of light due to the expansion of the universe, introduce additional complications. The redshift also tells us about the age of the universe at which we observe the galaxies The observed features of galaxies can be significantly altered by redshift, making it challenging to compare mergers at different distances and epochs. Moreover, the subjective nature of visual classification plays a significant role in the overall complexity of the task. Different observers may interpret the same features in varying ways, leading to inconsistent categorization outcomes. This subjectivity underscores the importance of developing advanced analytical techniques and employing comprehensive observational data in order to enhance the precision and reliability of galaxy merger classification. Such advancements are crucial for overcoming the multitude of challenges posed by the complex, dynamic, and multifaceted nature of galaxy mergers.

1.3 Differences in Real and Simulated Data

The figure 1.1 offers a detailed presentation of images of 10 carefully simulated galaxies, each meticulously viewed from three distinct and varied angles. This comprehensive approach contrasts starkly with the observation of real galaxies, where we are often constrained by a single viewing angle which does not allow us to accurately determine the history. The figure was generated following the same approach utilized in [1]. Unlike real galaxies, where our knowledge is limited to current observations, the full and detailed history of these simulated galaxies is completely known, enabling us to precisely determine whether they are destined to undergo future mergers. This complete historical knowledge is invaluable as it allows researchers to predict and analyze future interactions and mergers with a high degree of accuracy [5]. Moreover, the simulation process is accelerated, giving us the unique ability to observe and confirm potential mergers within a galaxy cluster without having to wait billions of years for the mergers to occur. This ability to speed up the simulation is crucial for studying long-term galactic evolution, which would otherwise take billions of years to observe naturally.

Additionally, the capacity to generate images from multiple distinct perspectives, as illustrated in the figure, provides an opportunity for a more comprehensive study of each galaxy's intrinsic and extrinsic properties. By analyzing galaxies from various angles, we can uncover details about their structure, morphology, and potential interactions that would otherwise remain hidden in a single view. The simulated data is obtained from the Illustris-TNG project [9]. This level of detail and flexibility, inherent to simulated galaxies, is simply not possible with real galaxy images, where observational constraints limit us to single-angle views, and we lack complete, precise information about their merger histories [1]. The inability to change the viewpoint or to access the full historical context in real galaxy observations significantly hampers our ability to fully understand their evolution and future interactions. Therefore, simulations provide a powerful tool to overcome these limitations, offering insights that are unattainable through traditional astronomical observations alone.

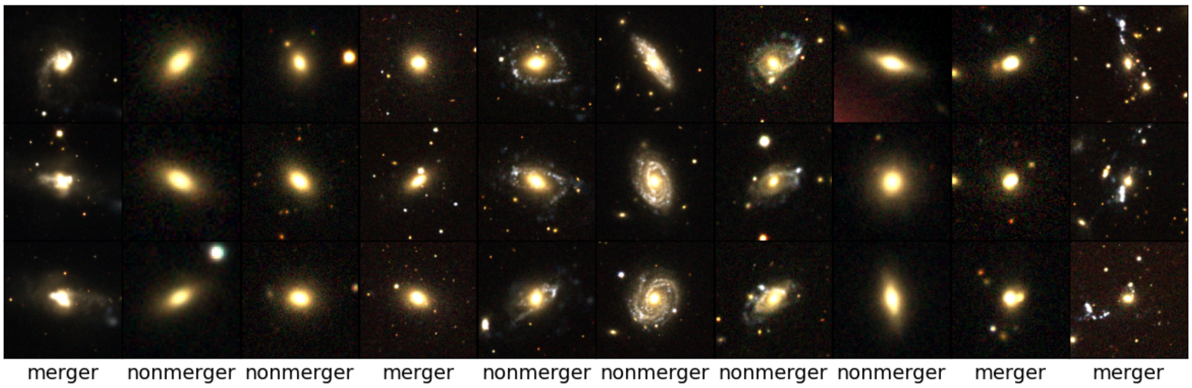


Figure 1.1: Images of 10 simulated galaxies from the Illustris-Tng project [1]

The figure 1.2 below presents a collection of 30 images, each depicting a different real galaxy, meticulously sourced from the Sloan Digital Sky Survey (SDSS) Database [10]. These images represent a significant achievement in astronomical observation, yet they come with inherent limitations due to the nature of the data. Because these images are taken from actual telescopic observations, we are confined to single-angle views, meaning that we only see each galaxy from one specific perspective. This restriction prevents us from gaining a complete understanding of the galaxies' structures and hinders our ability to fully grasp their complex morphologies. Moreover, we lack comprehensive details about their past merger histories and dynamical properties, which are critical for understanding their evolution. This limitation arises primarily because, apart from the spectroscopic data that provides information on the composition and velocity of galaxies and the visual data captured by the telescope, there is little additional information available. The data we possess is limited to what can be observed directly, making it exceedingly challenging to classify these galaxies accurately as mergers or non-mergers [7]. The absence of historical data on these galaxies' interactions further complicates efforts to understand their current state and predict their future evolution. To address these significant challenges, researchers must generate simulated galaxies that can replicate the observable properties of these real galaxies. By creating these virtual counterparts, scientists can systematically study their properties in ways that are not possible with real galaxies. Simulated galaxies allow us to explore different angles, analyze the effects of various initial conditions, and observe the dynamics of galaxy mergers over time, all of which are crucial for a more in-depth understanding. Through detailed analysis of these simulations, we can gain a better understanding of the interactions, dynamics, and evolutionary processes of real galaxies [11]. This approach not only enhances our knowledge of their physical characteristics but also provides insights into their merger histories, which are often obscured in real observations. By studying these virtual models, astronomers can draw parallels and make more informed interpretations of the data gathered from real galaxies, ultimately leading to deeper insights into the complex and dynamic nature of galaxy evolution [12].

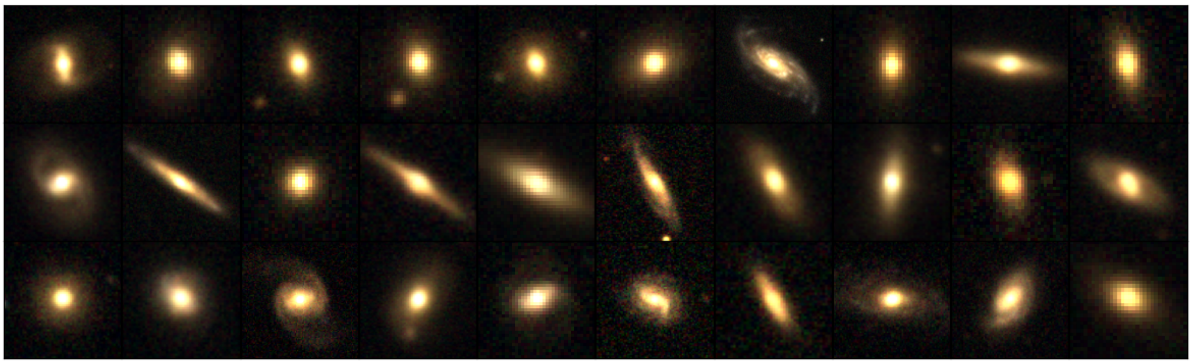


Figure 1.2: Images of 30 galaxies from the Sloan Digital Sky Survey [1]

1.4 Scattering Transform Networks

Utilizing scattering transforms for galaxy merger classification offers distinct and valuable advantages, particularly when dealing with complex astronomical data. One of the key strengths of scattering transforms is their inherent translation invariance, which ensures the model's robustness against variations in the observation angles of galaxies. This feature is especially crucial for galactic images, as they are often captured from different perspectives and orientations. The ability to maintain classification performance despite these variations makes scattering transforms well-suited for the task of identifying and analyzing galaxy mergers, where the structural appearance of galaxies can shift significantly based on the angle of observation [6]. Traditional methods might struggle with such variations, but the translation invariance of scattering transforms enables a consistent and reliable analysis across diverse observational data.

In addition to their translation invariance, scattering transforms excel in multiscale analysis [3]. Galaxy mergers typically involve intricate structures and features across a wide range of spatial scales, from the large-scale tidal tails and extended halos to the small-scale compact cores of interacting galaxies [8]. Scattering transforms, with their ability to capture information across multiple scales, are particularly adept at identifying and distinguishing these different features. This multiscale capability enhances the discrimination between various stages and types of galaxy mergers, which is essential for accurately classifying mergers into categories such as early-stage, late-stage, or post-merger systems. The ability to capture both fine details and broader structures allows for a more comprehensive understanding of the interaction dynamics.

The nonlinear operations embedded within scattering transforms also contribute to their strength in classification tasks. These operations are designed to extract highly discriminative features from the data, making them well-suited for pattern recognition tasks [6]. In the context of galaxy mergers, the complex and often chaotic nature of interacting galaxies can produce a wide variety of patterns that are difficult to differentiate using traditional linear methods [13]. Scattering transforms, by leveraging nonlinearities, are able to effectively capture and highlight these patterns, thus improving the classification accuracy of galaxy mergers. This capability becomes particularly beneficial when dealing with large datasets of galaxy images, where subtle differences between merger types or stages might otherwise be overlooked.

Another significant advantage of scattering transforms is their ability to facilitate dimensionality reduction without sacrificing important information. High-dimensional data, such as astronomical images, can be challenging to work with due to the computational demands and the risk of overfitting [13]. Scattering transforms address this issue by reducing the dimensionality of the data while preserving key features that are critical for classification. This reduction not only simplifies the computational complexity but also enhances the model's robustness to noise, which is a common issue in observational astronomy. Observational data, especially from distant galaxies, is often affected by various forms of noise, such as instrumental noise or atmospheric interference [7]. The robustness of scattering transforms to such noise ensures that the extracted features remain reliable, even in the presence of noisy data.

In essence, scattering transforms provide a powerful toolkit for galaxy merger classification, offering a blend of translation invariance, multiscale analysis, nonlinear feature extraction, dimensionality reduction, noise robustness, and interpretability [3]. These attributes make them an ideal choice for tackling the challenges posed by astronomical image data and for advancing our understanding of galaxy mergers.

Figure 1.3: Implementation of scattering transform on CIFAR data

Scattering transforms work by analyzing the edges and textures of the original image and calculating distances between the edges and features in the images being compared. In the context of galaxy merger classification, this process helps in identifying similarities and differences between images, allowing for more precise classification [6]. The figure 1.3 illustrates how scattering transforms can be applied to find images using the CIFAR [14] dataset similar to a given reference, such as a car. The figure shows scattering transforms applied using two different distance calculation methods: the L2 norm and the L1 norm. This example highlights the versatility of scattering transforms in capturing the relevant features of an image for comparison, which is directly applicable to galaxy merger analysis as well.

Left most image shows the original image. The second image shows the image found after the l2 norm calculation. The third image shows the image found after the implementation of the scattering transform on the l2 norm and the fourth shows the image found after implementing the scattering transform on the l1 norm. The result is not quite right as we can see a bird and a dog as 2 of the results. This is because the algorithm gets confused by the edges in the background of the image. This idea is chosen because we do not have detailed high-resolution images of real SDSS galaxies [10] and traditional CNNs or transformers will not work well with such low-resolution images [12, 8]. Scattering Transform on the other hand does not require high-quality images rather it would get misled by a high number of edges in a high-resolution image. Images of galaxies after background sky subtraction [1] is an ideal dataset for Scattering Transform to be implemented.

1.5 Objectives

This project aims to analyse the data after applying the scattering transform and obtain parameters of the image used to determine the merger properties. This process would be applied to both simulated and real images of mergers. The statistics after applying the scattering transform to both real and virtual images would be compared to determine the properties of galaxy mergers.

Goals to be met:

- Obtaining Simulated and real data for Galaxies
- Developing a Scattering transform network to be implemented on real and simulated images of galaxy mergers
- Understanding the application of Scattering transform networks to image datasets

Chapter 2

Literature and Technology Survey

2.1 Galaxy mergers and their significance

Galaxy mergers are collisions of galaxies. These events trigger gravitational interactions and friction between gas and dust, which significantly impact the involved galaxies. Understanding galaxy mergers is crucial for grasping galaxy evolution, yet accurately identifying such mergers remains a challenge [7]. The specific effects of mergers depend on various factors like collision angles, speeds, and relative size/composition. The merger rate, a fundamental measurement in galaxy evolution, offers insights into how galaxies have evolved over time and grown in mass. In the current Λ cold dark matter (ΛCDM) model of universe structure formation,

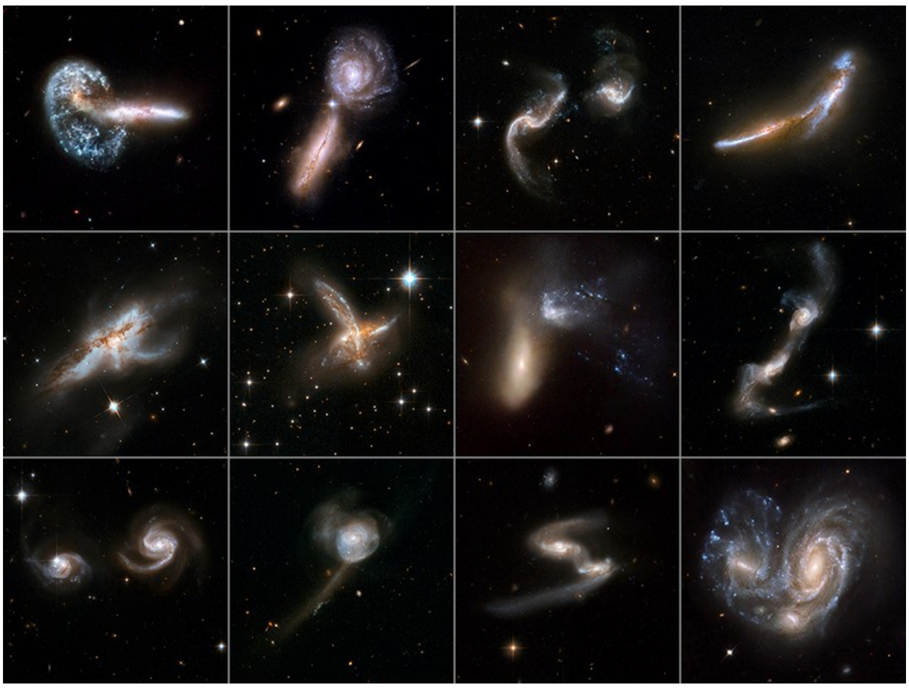


Figure 2.1: Images of Galaxy Mergers [2]

galaxies form as gas cools within dark matter halos' central regions [12]. This process has been detailed in numerous studies. Subsequently, galaxies grow through gas accretion and mergers, transitioning from small, irregular galaxies with high star formation rates to larger, quiescent galaxies with lower star formation rates. Supermassive black holes (SMBHs) at massive galaxies' cores also accumulate mass progressively through gas accretion. Active galactic nuclei (AGNs) represent SMBHs actively accreting gas, often displaying high luminosity. There's evidence suggesting a coevolution between SMBHs and their host galaxies, although the exact triggering mechanisms for AGNs and star formation remain uncertain[12]. Secular instabilities driven by disks and spiral arms have also been identified as significant factors in galaxy evolution.

However, many details of these processes, such as the timing and mechanisms involved in AGNs and galaxy evolution, remain unclear. A significant challenge is constructing an accurate observational sample of galaxy mergers, both major and minor.

Major mergers occur when two galaxies of comparable mass (typically within a mass ratio of 1:1 to 1:3) merge [5]. Major mergers have a significant impact on the structure of the resulting galaxy, often disrupting spiral galaxies and leading to the formation of elliptical galaxies. They can trigger starbursts and are thought to be an important mechanism in galaxy evolution.

Minor mergers occur when a large galaxy merges with a much smaller one, typically with a mass ratio greater than 3:1 [5]. Minor mergers are less disruptive than major mergers and tend to affect the morphology of the larger galaxy more subtly. However, they can still drive significant changes in star formation rates and contribute to the growth of the larger galaxy's mass.

Major mergers with specific mass ratios are believed to generate tidal torques that fuel gas accretion, star formation, and SMBH growth[7]. Conversely, minor mergers or continuous gas accretion are proposed to play critical roles in shaping galaxy morphologies and contributing to SMBH mass growth. Various methods like the Gini-M20 or CAS methods have been employed to identify mergers in imaging surveys. However, these methods have respond differently and are sensitive to different merger initial conditions and stages.

2.2 Scattering Transform

Scattering transform is a multiscale convolutional transform that produces a hierarchy offeatures[6, 13, 15]. The extracted features exhibit symmetries typically found in images, such as translation invariance. For that reason, they are commonly used in image processing. In the context of this project simulated and real images of galaxies are used and scattering transform is applied on these images. A Scattering transform has the structure of a convolutional network, but its filters are given by wavelets instead of being learned[16, 3]. Scattering representations construct invariant, stable, and informative signal representations by cascading wavelet modulus decompositions followed by a lowpass filter. A wavelet decomposition operator at scale J is defined as shown below.

$$W_\lambda x(u) = x \star \psi_\lambda(u) = \int x(v)\psi_\lambda(u-v)dv$$

where $\psi_\lambda(u) = 2^{-dj}\psi(2^{-j}r^{-1}u)$ and $\lambda = 2^j r$ with $j < J$ and $r \epsilon G$ belongs to a finite rotation group G of $\mathbb{R}^d$. We say, $\lambda \epsilon \Lambda j$ where $\Lambda j = \{2^j r : r \epsilon G/\{\pm 1\}, |\lambda| < 2^J\}$. Each rotated and dilated wavelet thus extracts the energy of x located at a given scale and orientation given by λ. As a result, note that the larger j is, the wider $\psi_\lambda(u)$ is. The difficulty is that the

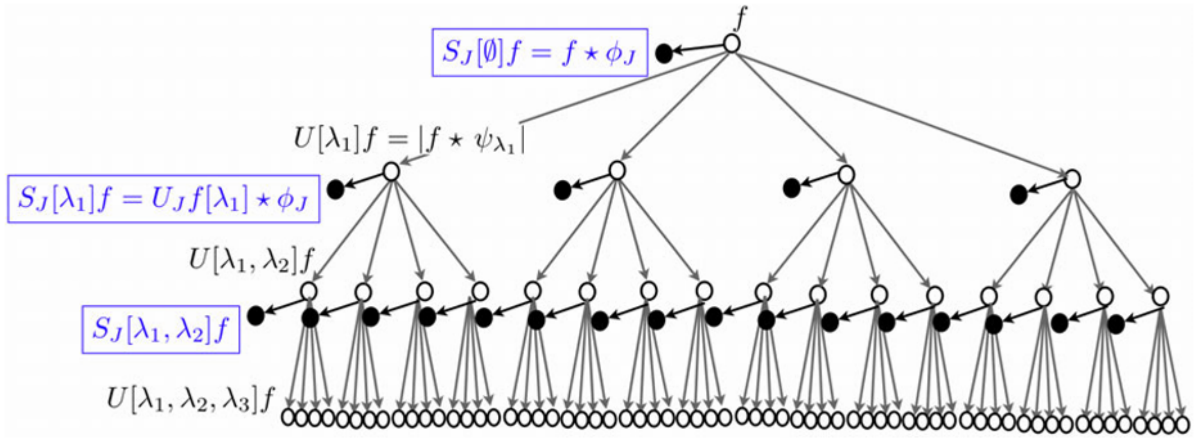


Figure 2.2: Scattering transform Network showcase [3]

wavelet coefficients are not translation invariant. A translation invariant representation can be obtained by averaging the coefficients over $\mathbb{R}^d$. However such averaging does not produce any information because wavelets have zero mean:

$$\int_{\mathbb{R}^d} (x * \psi_\lambda)(u)du = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} x(t)\psi_\lambda(u-t)dtdu = \int_{\mathbb{R}^d} x(t) \underbrace{\left(\int_{\mathbb{R}^d} \psi_\lambda(u-t)du\right)}_{0} dt = 0.$$

A translation invariant measure can be extracted out of each wavelet sub-band λ by introducing a non-linearity which restores a non-zero, informative average value. This is for instance achieved by computing the complex modulus and averaging the result

$$\int |x \star \psi_\lambda| du.$$

The information lost by this averaging is recovered by a new wavelet decomposition

$$|x \star \psi_\lambda| \star \psi_{\lambda}^{'} \lambda^{'} \epsilon \Lambda_J$$

of $|x \star \psi_\lambda|$, which produces new invariants by iterating the same procedure.
Let $U[\lambda]x = |x \star \psi_\lambda|$ denote the wavelet modulus operator corresponding to the subband λ. Any sequence $p = (\lambda_1, \lambda_2, ..., \lambda_m)$ defines a path, i.e., the ordered product of non-linear and noncommuting operators

$$U[p]x = U[\lambda_m] ... U[\lambda_2]U[\lambda_1]x = |||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \cdots \star \psi_{\lambda_m}|$$

with $U[0]x = x$.
Many applications in image and audio recognition require locally translation invariant representations, which also keep spatial or temporal information intact beyond a certain scale 2^J . A windowed scattering transform computes a locally translation invariant representation by

computing a lowpass average at scale 2^J with a lowpass filter $\phi_{2^J}(u) = 2^{-2^J}\phi(2^{-J}u)$. For each path $p = (\lambda_1, ..., \lambda_m)$ with $\lambda_i \epsilon \lambda_J$ the windowed scattering transform can be defined as

$$S_J[p]x(u) = U[p]x \star \phi_{2^J}(u) = \int U[p]x(v)\phi_{2^J}(u-v)dv.$$

A Scattering transform has the structure of a convolutional network, but its filters are given by wavelets instead of being learned. The figure 2.2 shows a representation of Scattering transform Network. These mathematical representations of scattering transform are taken from the following papers [3, 16]

2.3 Dataset

Simulated Data

The dataset for this project would comprise of real and simulated images of galaxies. We would also require images of real galaxies to compare them with their simulated counterparts. The plan is to use simulated images from the Illustris TNG project. The IllustrisTNG project is a series of large, cosmological magnetohydrodynamical simulations of galaxy formation. TNG aims to illuminate the physical processes that drive galaxy formation: to understand when and how galaxies evolve into the structures that are observed in the night sky, and to make predictions for current and future observational programs. The simulations use a state-of-the art numerical code which includes a comprehensive physical model and runs on some of the largest supercomputers in the world. IllustrisTNG is a suite of large volume, cosmological, gravo-magnetohydrodynamical simulations run with the moving-mesh code [9, 4] . The TNG project is made up of three simulation volumes: TNG50, TNG100, and TNG300, shown in figure 2.3 .

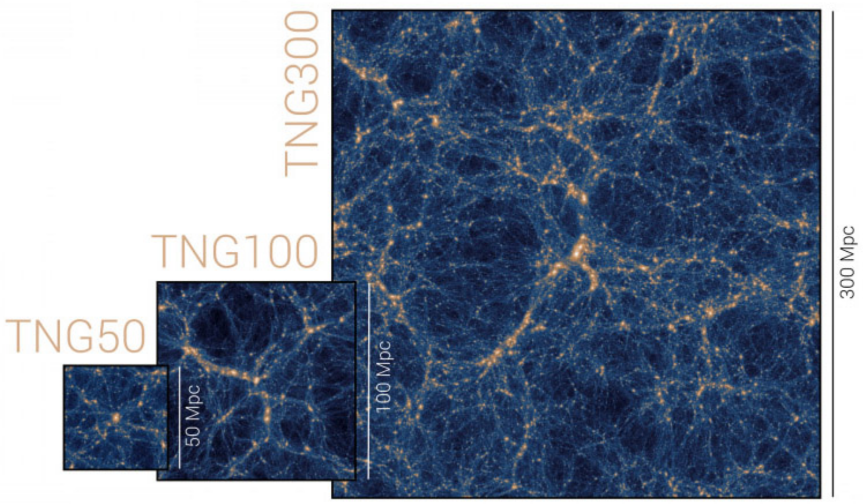


Figure 2.3: The three IllustrisTNG simulation volumes: TNG50, TNG100, and TNG300 [4]

The first two simulations, TNG100 and TNG300, were recently introduced in a series of five presentation papers [17, 18, 19, 20, 21] . The third and final simulation of the project is TNG50 [22, 23]. All three simulations are now publicly released. TNG includes a comprehensive model for galaxy formation physics, which is able to realistically follow the formation and evolution of galaxies across cosmic time [24, 25] . Each TNG simulation self-consistently solves the coupled evolution of dark matter, cosmic gas, luminous stars, and supermassive blackholes from a starting redshift of z=127 to the present day, z=0. The data is obtained and processed by following the methodology used in [1]

Real Data

Real data would be required to validate the effectiveness of the algorithm used and whether the algorithm could provide similar results compared to the simulated counterparts. The Sloan Digital Sky Survey (SDSS) is one of the most ambitious and influential surveys in the history of astronomy. Over eight years of operations (SDSS-I, 2000-2005; SDSS-II, 2005-2008), it obtained deep, multi-color images covering more than a quarter of the sky and created 3-dimensional maps containing more than 930,000 galaxies and more than 120,000 quasars. The SDSS used a dedicated 2.5-meter telescope at Apache Point Observatory, New Mexico, equipped with two powerful special-purpose instruments. Since its operations began in 1998, SDSS has been taking near-continuous observations of stars, galaxies, and quasars, and other objects, spanning from our solar system to the early days of the Universe [10]. The 120-megapixel camera imaged 1.5 square degrees of sky at a time, about eight times the area of the full moon. A pair of spectrographs fed by optical fibers measured spectra of (and hence distances to) more than 600 galaxies and quasars in a single observation. A custom-designed set of software pipelines kept pace with the enormous data flow from the telescope. The two key technologies that enabled the SDSS, optical fibers and the digital imaging detectors known as CCDs, were the discoveries awarded the 2009 Nobel Prize in Physics. There are numerous ways to access the SDSS DR18 data products, summarized on the SDSS website3 and on the data release website4. As in previous phases of SDSS, SDSS-V provides a searchable database for the Catalog Archive Server (CAS), using SkyServer5, with both SQL and Jupyter notebook interfaces. [10]

2.4 Astropy

Python is a high-level, interpreted programming language that has quickly become a standard in various computational domains, technological sectors, and research fields. Its rapid adoption is due to its ability to execute scalable, time- and energy-efficient code while emphasizing readability, ease of use, and seamless integration with other languages. In the past decade, Python's popularity has surged, making it a leading programming language, particularly within the astronomical and broader scientific communities [26]. The astropy core package was one of the pioneering large-scale, open-source Python libraries created for astronomy. It offers a range of functionalities, including tools for reading and writing astronomy-specific data formats (such as FITS), handling and converting astronomical coordinates and managing and propagating physical units within code. There are multiple string representations for units used in the astronomy community. A standard astronomical example is the relationships between the frequency, wavelength and energy of a photon – it is common practice to treat such units as equivalent even though they are not strictly comparable. Such a conversion can be carried out in astropy.units by supplying an equivalency list [27].

The flexible image transport system (FITS) standard [28] defines a unit standard, as well as both the Centre de Données astronomiques de Strasbourg [29] and NASA/Goddard's Office of

Guest Investigator Programs[30]. In addition, the International Virtual Observatory Alliance (IVOA) has a forthcoming VOUnit standard [31] in an attempt to resolve some of these differences. Rather than choosing one of these, astropy.units support most of these standards.

The images from the datasets mentioned both SDSS and Illustris-TNG have their data available in FITS extension format. Astropy is used as a library in this project with python to read these FITS files.

Chapter 3

Methodology

3.1 Introduction

Highlighted in chapters 1 and 2 the aim of this project is to ascertain properties of galaxy mergers by using scattering transform [3] on real (SDSS) [10] and simulated (Illustris-TNG) [9] images of galaxies. This is done by first loading data from Illustris-TNG and Sloan Digital Sky Survey for Simulated and Real images. Around a total of 4654 Illustris-TNG files are used. Every TNG is a FITS file which represents the data of a simulated galaxy [28]. Since these are simulated data files each file can be used to view the galaxy from multiple angles. Each TNG FITS file is used to view the galaxy from 3 different angles in this project. The three views are saved as 3 separate image files. Hence a total of 13,959 simulated images are used. A total of 16,981 SDSS files are used for real images. Every SDSS file is in FITS format.

Astropy is used for reading the TNG and SDSS data files as mentioned in chapter 2 [27]. Astropy allows use of python to view the images and also read the metadata associated with each FITS file. The metadata consists of important details about the image like the redshift of the object in the image, details of spectra, stellar mass, class of the object and many more intricate details. Figure 1.1 and 1.2 are a few images acquired after processing the FITS files for simulated and real data respectively.

Kymatio is a Python library used to implement Scattering Transform. Scattering Transform is implemented with the help of this library on the real and simulated data [32]. As a result, the best distance for each real data point is calculated. Kymatio documentation page comes with a few examples on how to implement the Scattering transform algorithm. This page was instrumental in understanding the usage of the library. CIFAR dataset [14] was used as proof of concept as shown in Figure 1.3.

A web application is developed to visualize and analyse the results obtained after the implementation of the Scattering transform algorithm on the data. The web application makes it easier for a researcher to infer details about the matched galaxies and how they compare to the original image. The web application is deployed on the Heroku cloud cluster.

3.2 Setting up code environment

The code is designed to calculate distances for 16,981 SDSS [10] images, which are first transformed into tensors for computation. Given the large number of image tensors, this process demands a GPU with sufficient memory to support tensor operations. For this project, an NVIDIA RTX 3060 with 13GB of VRAM is used to handle the computational load. The CUDA library is installed to enable GPU acceleration, optimizing performance. PyTorch is utilized to facilitate the conversion of images into tensors. The coding environment includes Jupyter Notebook, integrated into Visual Studio Code (VSCode), which serves as the IDE for developing and executing the project.

3.3 Image preprocessing

FIrst the FITS data files are loaded using the astropy library [27]. Every FITS data point is converted into 3 PNG images for simulated data and 1 PNG image for real data. A sky subtraction is performed on every image before saving them to get rid of the background noise, similar to the approach utilized in [1]. This is done to avoid errors in the Scattering Transform algorithm since it can misread edges in the background sky of the object. These png files are then loaded again and processed before passing it through the Scattering Transform function. The figure 3.1 shows how Sky subtraction is performed in every image. As seen in the figure the object in the image is more clearly visible after sky subtraction.

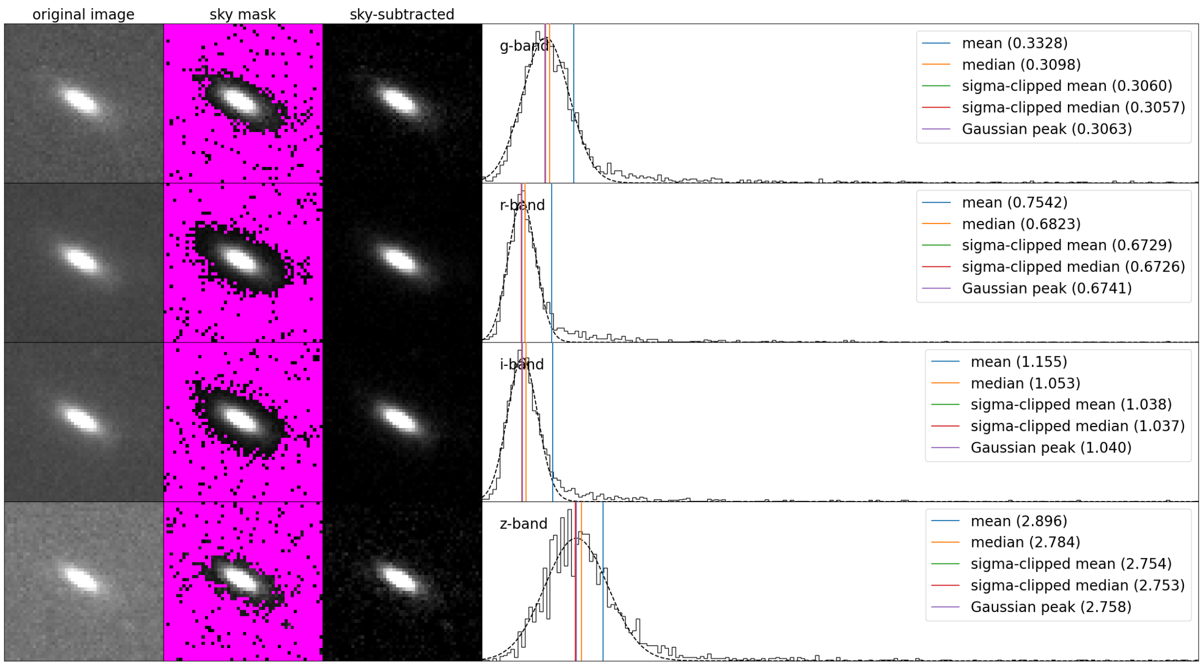


Figure 3.1: Sky subtraction implementation [1]

Pytorch is used to normalise the images and transform them into tensors. These images are resized into 32 x 32 sized images with 3 channels(rgb) These tensors are passed to the Scattering2D function.

3.4 Implementing Scattering Transform

A Scattering2D object is created using the Kymatio library [32]. This object is used later to implement Scattering. The code is designed to identify the closest matches between images from two datasets, specifically `sdss_ims` and `tng_ims`, using a variety of distance metrics. Here `sdss_ims` is a list of tensors which is generated by transforming the SDSS(Real) images of galaxies [10]. Similarly, `tng_ims` is also a list of tensors which is generated by transforming the Illustris-TNG(Simulated) [9] images of galaxies. The process unfolds through several key steps:

L1 Norm:

The L1 norm [33] of a vector is the sum of the absolute values of its components. For a vector $x = (x_1, x_2, ..., x_n)$, the L1 norm is calculated as: $L1(x) = |x_1| + |x_2| + ... + |x_n|$. The L1 norm measures the distance between two points as the sum of the absolute differences of their coordinates. This is often referred to as "Manhattan distance" because it's like finding the shortest distance between two points in a city grid where you can only travel along streets. L1 norm tends to promote sparsity, meaning it encourages many elements of the vector to be zero. This is useful in feature selection and regularization techniques.

L2 Norm:

The L2 norm [33] of a vector is the square root of the sum of the squares of its components. For a vector $x = (x_1, x_2, ..., x_n)$, the L2 norm is calculated as: $L2(x) = \sqrt{{x_1}^2 + {x_2}^2 + ... + {x_n}^2}$. The L2 norm measures the straight-line distance between two points. This is also known as "Euclidean distance." The L2 norm is more sensitive to outliers. Large values in a vector can significantly impact the L2 norm.

Initialization:

The code begins by initializing three empty lists, `best_l2_indices`, `best_sc_l2_inds`, and `best_sc_l1_inds` to store the indices of the closest matching images for each image in `sdss_ims`. Next, `ttng_ims` is generated by concatenating the images in `tng_ims` and transferring them to the specified device, such as a GPU, for accelerated computation. Similarly, `scttest_ims` is created by applying a function, `sc`, to `ttng_ims` and moving the resulting transformed images to the same device.

Loop through `sdss_ims`:

The code then iterates over each image, `im`, in `sdss_ims`. During each iteration, the image is first transferred to the specified device. A transformed version of the image, `scim`, is created by applying the `sc` function to `im` and similarly transferring the result to the device.

L2 Distance Calculation:

The L2 distance [33] between `im` and each image in `ttng_ims` is calculated as the sum of squared differences across all dimensions except the batch dimension. The indices of `ttng_ims` are then sorted based on their L2 distance to `im`, and the top 5 closest matches are appended to the `best_l2_indices` list.

Scattering Transform L2 Distance Calculation:

The code then calculates the L2 distance [33] between `scim` and each image in `scttest_ims`, which are the scattering transformed versions of the images [6, 3]. These distances are calculated similarly, to the sum of squared differences across all relevant dimensions. The indices of `scttest_ims` are sorted according to their L2 distance from `scim`, and the closest 5 matches are added to the `best_sc_l2_inds` list.

Scattering Transform L1 Distance Calculation:

Finally, the L1 distance [6] is computed between `scim` and each image in `scttest_ims`, calculated as the sum of absolute differences across all relevant dimensions [6, 3]. The indices of `scttest_ims` are sorted by their L1 distance from `scim`, and the top 5 closest matches are appended to the `best_sc_l1_inds` list.

Saving the Distances and normalised dataset for use later:

The `pickle` module to serialize and save several variables to disk. It first opens a binary write file named `best_l2_indices.pkl` and dumps the contents of the variable `best_l2_indices` into it. This process is repeated for the variables `best_sc_l2_inds`, `best_sc_l1_inds`, `tng_ims`, `sdss_ims` and `tng_loader`, each being saved to their respective `.pkl` files. This allows these variables to be easily loaded back into Python code later using the `pickle.load` function, preserving their data structure and content. `sdss_loader` needs to be generated again in the code where it needs to be used because it was generated using a custom dataloader function unlike the tng_loader.

3.5 Storing results for analysis:

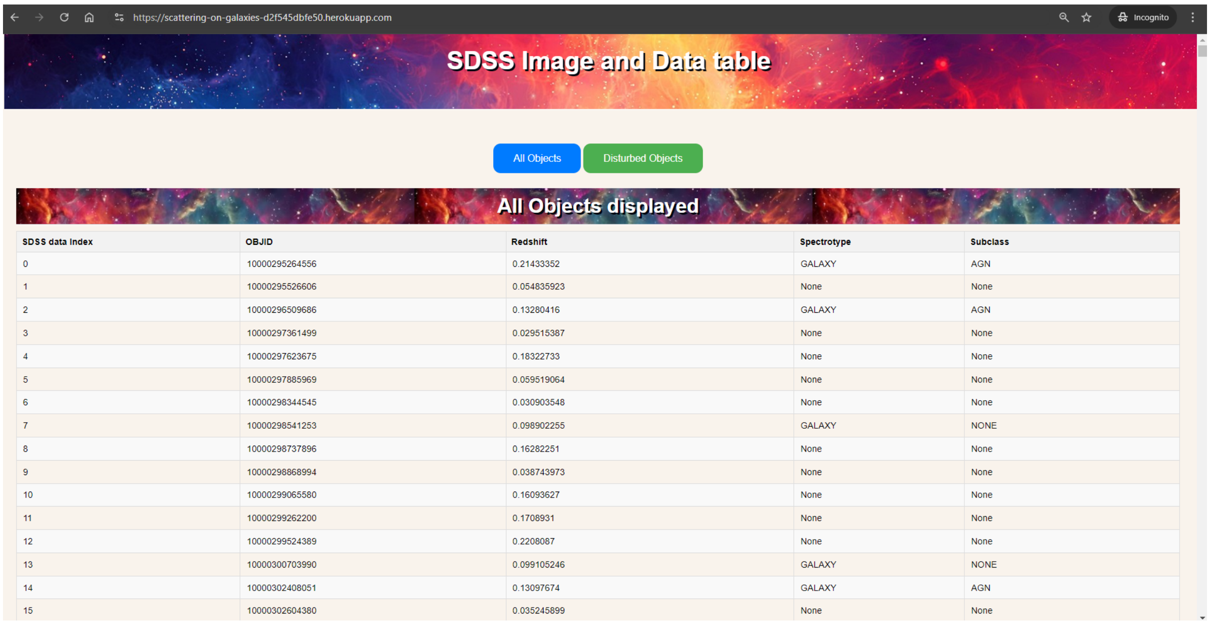
Every .fits object has its own respective metadata which can be obtained by using the fits.getheader function. However, displaying the resulting images obtained after applying scattering Transform with their respective headers for each sdss image in the dataset proves to be a challenging task. A more easier method to view results was required for the purpose of this project. Hence a web app using flask was developed and hosted for this purpose. To use the results and input data for the web app, a database named was created to store them using sqlite3. This process involved converting the image into Bytes and storing it alongside its respective header data in a single row of the table.

3.6 Flask web application:

A Flask web application is implemented for analyzing and visualizing galaxy data from both observational (SDSS) [10] and simulated (TNG) [9] datasets. The application connects to a SQLite database containing galaxy data. It provides two main routes: an index page that displays a list of galaxies (with an optional filter for disturbed galaxies) and a detailed view of individual galaxies.

Galaxy Data (scattering-on-galaxies-d2f545dbfe50.herokuapp.com)

The index route queries the database for galaxy properties such as stellar mass, redshift, and spectral type. It can filter results to show only disturbed galaxies if requested. The detailed view route retrieves comprehensive information about a specific galaxy, including its image and various morphological classifications.



SDSS data Index	OBJID	Redshift	Spectrotype	Subclass
0	10000295264556	0.21433352	GALAXY	AGN
1	10000295526606	0.054835923	None	None
2	10000296509686	0.13280416	GALAXY	AGN
3	10000297361499	0.029515387	None	None
4	10000297623675	0.18322733	None	None
5	10000297885969	0.059519064	None	None
6	10000298344545	0.030903548	None	None
7	10000298541253	0.098902255	GALAXY	NONE
8	10000298737896	0.16282251	None	None
9	10000298868994	0.038743973	None	None
10	10000299065580	0.16093627	None	None
11	10000299262200	0.1708931	None	None
12	10000299524389	0.2208087	None	None
13	10000300703990	0.099105246	GALAXY	NONE
14	10000302408051	0.13097674	GALAXY	AGN
15	10000302604380	0.035245899	None	None

Figure 3.2: Website Home page

A significant feature of this application is its image processing and visualization capabilities. It converts stored binary image data into numpy arrays, which are then rendered as images using matplotlib. These images are converted to base64-encoded strings for embedding in HTML responses. For each selected galaxy, it retrieves and displays similar galaxies from the TNG simulation dataset. This similarity is determined using pre-computed indices stored in pickle files, which represent the results of distance metrics (L2, scatter L2, and scatter L1) between the observational and simulated data.

This code structure allows for an interactive exploration of galaxy data, enabling users to view both observational data and comparable simulated galaxies. Such a tool is valuable for researchers studying galaxy morphology, evolution, and the relationship between observational data and theoretical models.

The website as shown in figure 3.2 displays a table of SDSS objects when it loads where each row is clickable. There are two buttons at the top of the page. Clicking these buttons can help one to toggle between all SDSS data objects used for this project and objects which have Disturbed sources. When one of the rows is clicked a new page 3.3 opens up which displays the SDSS image corresponding to that index. This page also enables viewing of the matched TNG objects and their corresponding data by scrolling below. As seen in the figure 3.4 one can toggle between the three matching methods and view the corresponding results. Scattering on L2 Norm is selected by default.

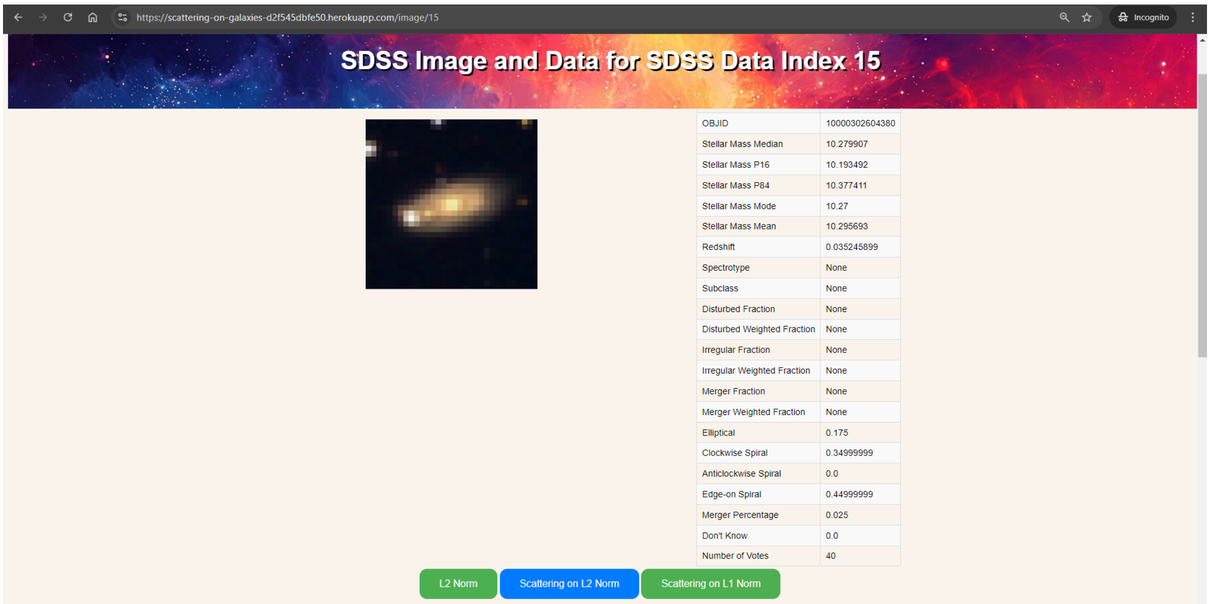


Figure 3.3: Data page part 1

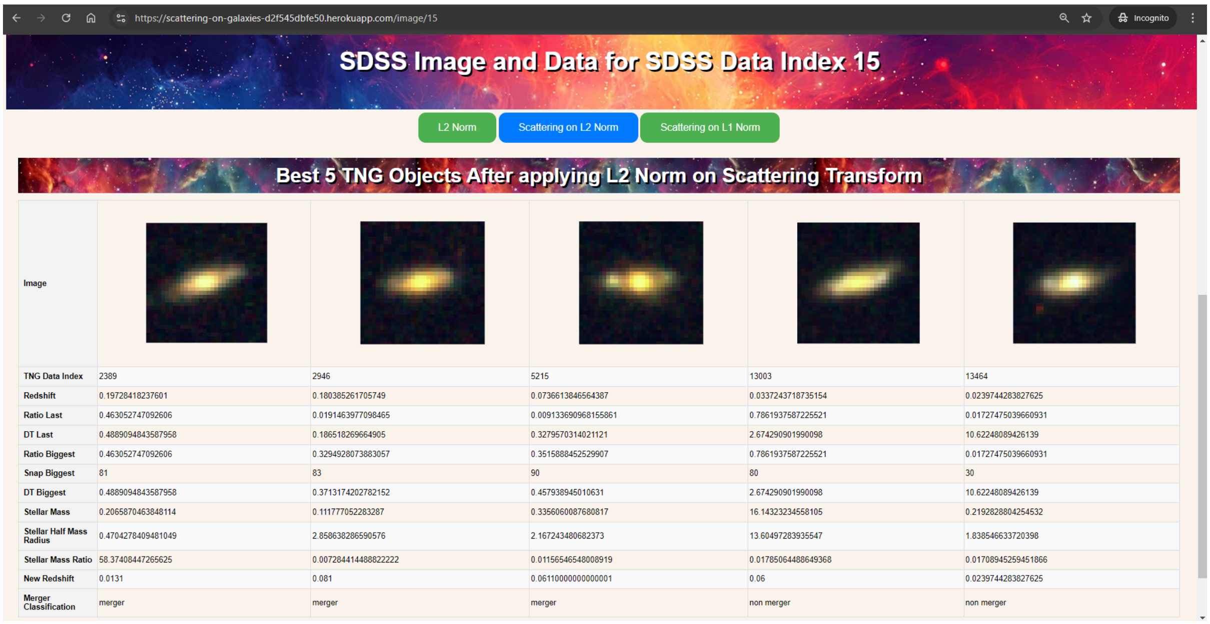


Figure 3.4: Data page part 2

3.7 Summary:

This research project presents an innovative approach to analyzing and comparing real and simulated galaxy images using scattering transform techniques. The study utilizes observational data from the Sloan Digital Sky Survey (SDSS) [10] and simulated data from the Illustris-TNG Project [9], comprising 16,981 real and 13,959 simulated galaxy images, respectively.

The methodology involves a sophisticated image processing pipeline. Initially, FITS files are processed using Astropy [27], with sky subtraction performed to minimize background noise.

The images are then normalized and converted to tensors using PyTorch, and resized to 32x32 pixel RGB images. The core of the analysis employs the scattering transform, implemented via the Kymatio library [32], to generate comparable representations of both real and simulated galaxies.

A key aspect of the study is the application of multiple distance metrics (L1 and L2 norms) [33] to quantify similarities between the scattering transform representations of real and simulated galaxies. This approach enables the identification of the most similar simulated galaxies for each observed galaxy, potentially providing insights into galaxy formation, evolution, and morphological characteristics.

To facilitate data exploration and analysis, a Flask-based web application was developed. This tool interfaces with a SQLite database containing the processed galaxy data, allowing for interactive visualization of galaxy images, their properties, and their closest simulated counterparts. The application renders images using matplotlib and employs base64 encoding for efficient web display.

This comprehensive approach, combining advanced image processing techniques [1], machine learning methods [6, 3], and interactive visualization tools represents a significant advancement in the comparative analysis of observational [10] and simulated [9] astronomical data. The developed framework has the potential to enhance our understanding of galaxy properties and merger events, bridging the gap between theoretical models and observational astronomy.

Chapter 4

Analysis and Results

The results have been obtained on following architecture:

- CPU: Intel I7 12700F
- GPU: Nvidea RTX 3060 13 GB VRAM
- System Memory 16GB RAM
- WSL: Ubuntu-22.04

To compare the TNG objects [9] to the original SDSS image [10], we would need to perform following tasks:

- **Visual inspection:** Directly compare the images to identify morphological similarities and differences.
- **Quantitative analysis:** Measure specific morphological features, such as ellipticity, concentration, and asymmetry, for both the TNG objects [9] and the SDSS image [10].
- **Feature-based comparison:** Use feature extraction techniques to identify common features between the images and quantify their similarity.

4.1 Analysis of SDSS object 1127605654002O845

Analysis of the SDSS image

The image 4.1 and data 4.1 represent observations of an elliptical galaxy.

1. Image: The image shows a bright, elliptical-shaped galaxy with a clearly defined core. The brightness decreases outwards from the center, which is typical for elliptical galaxies.
2. Object Classification:
   - Spectrotype: GALAXY
   - Subclass: AGN (Active Galactic Nucleus), indicating the presence of an active supermassive black hole at the center.
3. Stellar Mass: The galaxy has a high stellar mass, with median, mode, and mean values. This indicates it's a massive galaxy.

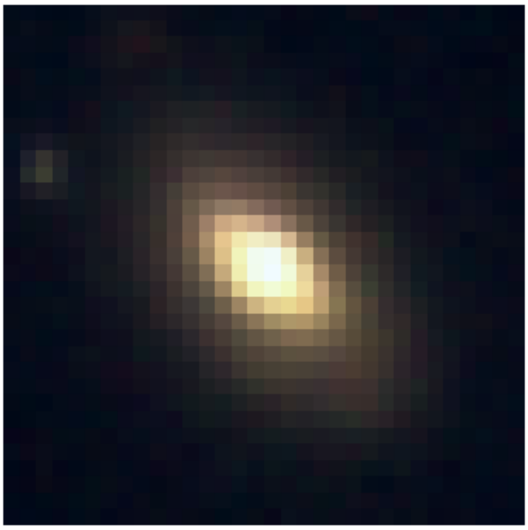


Figure 4.1: SDSS object 11276056540020845

Field	Value
OBJID	11276056540020845
Stellar Mass Median	10.683173
Stellar Mass P16	10.593278
Stellar Mass P84	10.777252
Stellar Mass Mode	10.690001
Stellar Mass Mean	10.696529
Redshift	0.062532574000000001
Spectrotype	GALAXY
Subclass	AGN
Disturbed Fraction	0.333
Disturbed Weighted Fraction	0.333
Irregular Fraction	0.333
Irregular Weighted Fraction	0.333
Merger Fraction	0.333
Merger Weighted Fraction	0.333
Elliptical	0.69999999
Clockwise Spiral	0.025
Anticlockwise Spiral	0.050000001
Edge-on Spiral	0.22499999
Merger Percentage	0.0
Don't Know	0.0
Number of Votes	40

Table 4.1: Data for SDSS object 11276056540020845

4. Redshift: 0.0625, suggesting it's relatively nearby in cosmological terms.

5. Morphology:
   - Elliptical: 0.69999999 (70%) probability
   - Edge-on Spiral: 0.22499999 (22.5%) probability
   - Small probabilities for clockwise and anticlockwise spirals
6. Disturbance Indicators: Disturbed Fraction, Irregular Fraction, and Merger Fraction are all 0.333, suggesting some level of disturbance or irregularity.
7. Merger: Despite the disturbance indicators, the Merger Percentage is 0, indicating that voters didn't classify it as a current merger.
8. Confidence: With 40 votes and no "Don't Know" responses, there seems to be good confidence in the classifications.

This appears to be a massive elliptical galaxy with an active galactic nucleus. While it shows some signs of disturbance or irregularity, it's not currently undergoing a merger. Its high mass and AGN classification suggest it's likely an evolved galaxy, possibly the result of past mergers [8], now in a relatively stable state but still showing some remnants of past interactions.

L2 Norm matched TNG objects and their analysis

This analysis explores the characteristics of the top 5 TNG (IllustrisTNG) [9] objects that exhibit the highest similarity to a reference SDSS (Sloan Digital Sky Survey) [10] image when measured using the L2 norm [33]. The L2 norm, also known as Euclidean distance, calculates the distance between two points in a multi-dimensional space. In this context, it measures the similarity between the pixel intensities of the TNG and SDSS images. The table4.2 shows the 5 objects and their respective details.

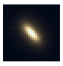
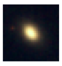
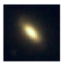
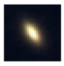
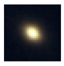
	Best 5 TNG Objects After applying L2 Norm				
Image					
TNG Data Index	1207	1247	5821	10815	11522
Redshift	0.243540181554671	0.243540181554671	0.0485236299818059	0.125759332411261	0.0994018026302219
Ratio Last	0.346932351589203	0.362385451793671	0.774995727115586	0.173803552985191	0.0325207747519016
DT Last	0.180094048669731	0.180094048669731	0.332136133304583	2.13404630932369	5.33410351364866
Ratio Biggest	0.346932351589203	0.362385451793671	0.774995727115586	0.322104096412659	0.0325207747519016
Snap Biggest	80	80	93	75	58
DT Biggest	0.180094048669731	0.180094048669731	0.332136133304583	2.30552678531926	5.33410351364866
Stellar Mass	9.47892665863037	3.76749753952026	0.282783865928649	23.5892687115478	6.63096284866333
Stellar Half Mass Radius	5.85940694808961	2.37988114356994	1.67968583106994	9.32988739013672	6.91591167449951
Stellar Mass Ratio	0.0373030155897141	0.0314746908843517	0.0118963774293661	0.314028799533844	0.112317949533463
New Redshift	0.1697	0.0672000000000001	0.0472000000000001	0.125759332411261	0.0994018026302219
Merger Classification	merger	merger	merger	non merger	non merger

Table 4.2: L2 Norm TNG matches for SDSS object 11276056540020845

Key Observations:

1. **Stellar Mass and Ratio:** The stellar mass varies significantly among the objects, indicating different stages of stellar evolution and potential interactions. The mass ratio represents the mass ratio of mergers.
2. **Merger History:** Three of the five objects have undergone mergers, suggesting that mergers are a common process in galaxy evolution.

Matched TNG objects for L2 Norm applied on Scattering transform

This analysis explores the characteristics of the top 5 TNG (IllustrisTNG) [9] objects that exhibit the highest similarity to a reference SDSS (Sloan Digital Sky Survey) [10] image when measured using the scattering transform with L2 Norm as distance calculation metric. The scattering transform [3] is a powerful tool for extracting invariant features from images, making it suitable for comparing objects with varying orientations and scales. The table4.3 shows the 5 objects and their respective details.

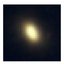
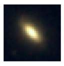
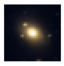
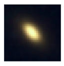
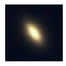
	Best 5 TNG Objects After applying L2 Norm on Scattering Transform				
Image					
TNG Data Index	1907	5821	10398	10642	10815
Redshift	0.214425035514495	0.0485236299818059	0.152748768902381	0.141876203969562	0.125759332411261
Ratio Last	0.879554349488888	0.774995727115586	0.321487933397293	0.0492636353763199	0.173803552985191
DT Last	0.304110333745486	0.332136133304583	8.69322298685802	8.20513049673549	2.13404630932369
Ratio Biggest	0.879554349488888	0.774995727115586	0.321487933397293	0.759434342384338	0.322104096412659
Snap Biggest	81	93	32	34	75
DT Biggest	0.304110333745486	0.332136133304583	8.69322298685802	8.50205702940323	2.30552678531926
Stellar Mass	12.0488138198853	0.282783865928649	1.21869552135468	0.104563534259796	23.5892687115478
Stellar Half Mass Radius	9.26597690582275	1.67968583106994	2.25735139846802	2.84451079368591	9.32988739013672
Stellar Mass Ratio	0.0323292165994644	0.0118963774293661	0.162302523851394	0.0211646854877472	0.314028799533844
New Redshift	0.214425035514495	0.0472000000000001	0.0637	0.0806	0.125759332411261
Merger Classification	merger	merger	non merger	non merger	non merger

Table 4.3: Scattering on L2 Norm TNG matches for SDSS object 11276056540020845

Key Observations:

1. **Redshift Distribution:** The redshift values range from 0.0334 to 0.1769, indicating a slightly lower redshift range compared to the previous set of objects.
2. **Stellar Mass and Ratio:** The stellar mass and stellar mass ratio vary across the objects, suggesting different evolutionary stages and potential interactions.
3. **Merger History:** Four of the five objects have undergone mergers, highlighting the prevalence of mergers in galaxy evolution.

Matched TNG objects for L1 Norm applied on Scattering transform

This analysis explores the characteristics of the top 5 TNG (IllustrisTNG) [9] objects that exhibit the highest similarity to a reference SDSS (Sloan Digital Sky Survey) [10] image when measured using the scattering transform with the L1 norm [33]. The table4.4 shows the 5 objects and their respective details.

Key Observations:

1. **Stellar Mass and Ratio:** The stellar mass varies significantly among the objects, indicating different stages of stellar evolution and potential interactions. the mass ratio represent sthe mass ratio of the mergers
2. **Merger History:** Three of the five objects have undergone mergers, suggesting that mergers are a common process in galaxy evolution.

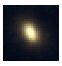
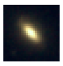
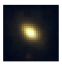
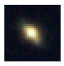
	Best 5 TNG Objects After applying L1 Norm on Scattering Transform				
Image					
TNG Data Index	1907	5821	11235	11602	12410
Redshift	0.214425035514495	0.0485236299818059	0.109869940458825	0.0994018026302219	0.0585073227945131
Ratio Last	0.879554349488879	0.774995727115586	0.0690980611741543	0.0100574912503362	0.541563749313354
DT Last	0.304110333745486	0.332136133304583	2.32770552300863	10.7758025866767	5.53955580918486
Ratio Biggest	0.879554349488879	0.774995727115586	0.0690980611741543	0.0123351616784931	0.541563749313354
Snap Biggest	81	93	76	19	60
DT Biggest	0.304110333745486	0.332136133304583	2.32770552300863	11.0981545412792	5.53955580918486
Stellar Mass	12.0488138198853	0.282783865928649	8.82521820068359	0.802594423294067	18.8179016113281
Stellar Half Mass Radius	9.26597690582275	1.67968583106994	4.00465297698975	1.75299096107483	7.61160755157471
Stellar Mass Ratio	0.0323292165994644	0.0118963774293661	0.0178031679242849	0.0314497786681259	0.0221393834799528
New Redshift	0.214425035514495	0.0472000000000001	0.109869940458825	0.0492000000000001	0.0585073227945131
Merger Classification	merger	merger	non merger	non merger	non merger

Table 4.4: Scattering on L1 Norm TNG matches for SDSS object 11276056540020845

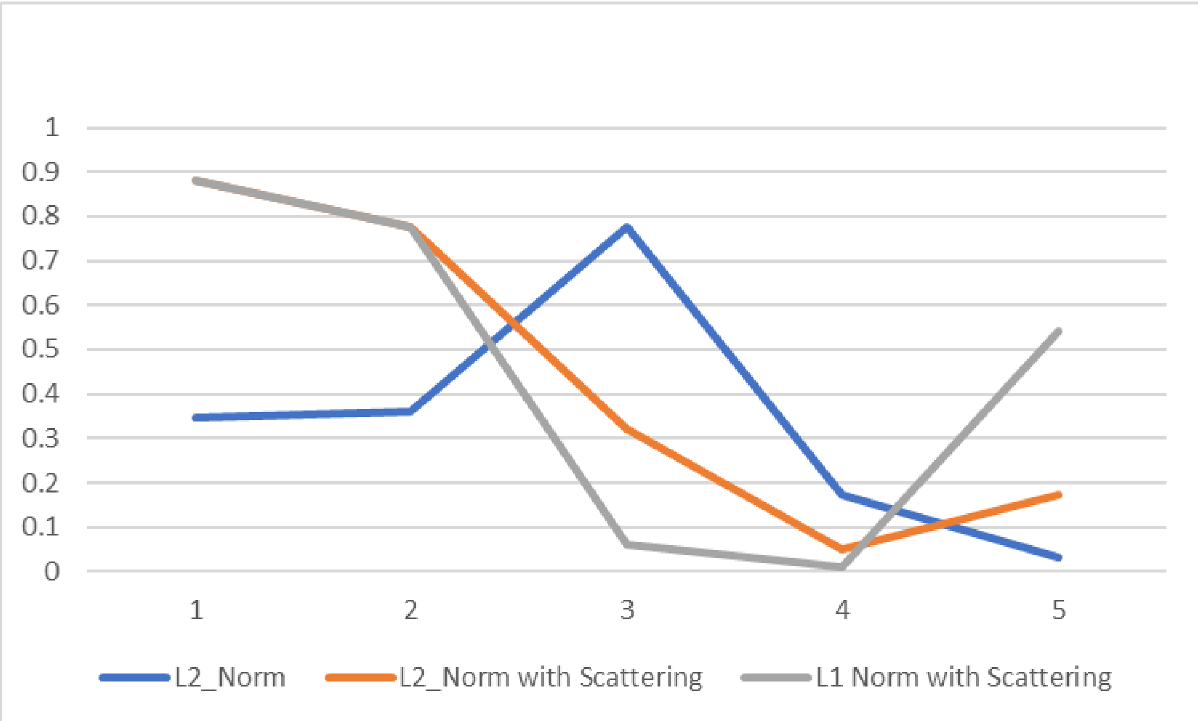


Figure 4.2: Ratio Last Comparison for matched TNG objects

Implications from comparisions

The "Ratio Last" values 4.2 represent the mass ratios of most recent mergers. Every match above 0.3 would be considered a major merger, so the closest matches across the 3 methods are quite similar. The ranking of the objects based on "Ratio Last" is not consistent across all metrics, suggesting that the metrics prioritize different aspects of similarity [22]. The scattering transform, both with L2 and L1 norms, seems to select objects with a wider range of "Ratio Last" values compared to the L2 norm alone. This might be due to the scattering transform's ability to capture more detailed morphological features. The "Ratio Last" metric provides insights into the distribution of dark matter within the objects, which is a crucial aspect of galaxy formation and evolution. The scattering transform's sensitivity to morphological features suggests that the distribution of dark matter and stellar mass is influenced by the overall structure of the galaxies.

While the top 5 objects selected by each metric may vary slightly, there is a degree of overlap, indicating that the metrics capture similar, but not identical, aspects of image similarity. The redshift and stellar mass distributions of the selected objects are consistent across all metrics, suggesting that these properties are fundamental to the similarity with the SDSS image. The merger history of the objects also appears to be a factor in their selection, with a majority of the objects having undergone mergers. The scattering transform metrics, both L2 and L1, are more sensitive to morphological features, as they capture invariant properties of the images. This might explain why the objects selected by these metrics may have more pronounced structural similarities to the SDSS image.

4.2 Analysis of SDSS object 10001374839131

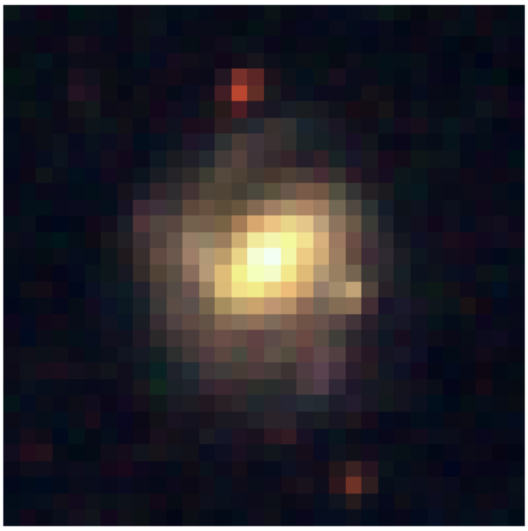


Figure 4.3: SDSS object 10001374839131

Analysis of the SDSS image

The image 4.3 of the galaxy has a bright central region with some evidence of spiral structure, as seen from the patterns of light surrounding the central core. The galaxy does not show any strong signs of disturbance or irregularity, indicating that it is a well-formed spiral galaxy.

The table provides various metrics and properties related to the galaxy:

1. **Stellar Mass:**
    - **Median:** 11.129943
    - **P16:** 11.02957

Field	Value
OBJID	10001374839131
Stellar Mass Median	11.129943
Stellar Mass P16	11.02957
Stellar Mass P84	11.223755
Stellar Mass Mode	11.133333
Stellar Mass Mean	11.140248
Redshift	0.14109947
Spectrotype	None
Subclass	None
Disturbed Fraction	None
Disturbed Weighted Fraction	None
Irregular Fraction	None
Irregular Weighted Fraction	None
Merger Fraction	None
Merger Weighted Fraction	None
Elliptical	0.0
Clockwise Spiral	0.97100002
Anticlockwise Spiral	0.028999999
Edge-on Spiral	0.0
Merger Percentage	0.0
Don't Know	0.0
Number of Votes	34

Table 4.5: Data for SDSS object 10001374839131

- **P84:** 11.223755
- **Mode:** 11.133333
- **Mean:** 11.140248

These values represent the galaxy's stellar mass in logarithmic units (likely in solar masses). The median and mean values are around 11.13, indicating that this galaxy has a significant stellar mass, consistent with large spiral galaxies.

2. **Redshift: Value:** 0.14109947 The redshift value indicates that this galaxy is relatively nearby in cosmic terms. A redshift of 0.14 places it at a moderate distance, possibly a few billion light-years away.

3. **Spectrotype/Subclass:** Both are listed as "None," meaning no specific spectral classification has been assigned in this dataset.

4. **Disturbance and Irregularity Metrics:**
   - **Disturbed Fraction/Weighted Fraction:** None
   - **Irregular Fraction/Weighted Fraction:** None
   - **Merger Fraction/Weighted Fraction:** None

These values suggest that there is no significant disturbance, irregularity, or merger activity in this galaxy, supporting the visual impression that it is a relatively undisturbed spiral galaxy.

5. **Morphology (Elliptical vs. Spiral):**
    - **Elliptical:** 0.0
    - **Clockwise Spiral:** 0.97100002
    - **Anticlockwise Spiral:** 0.02899999
    - **Edge-on Spiral:** 0.0

   The galaxy is overwhelmingly classified as a clockwise spiral (with a probability of 97%). There is a very small probability (approximately 3%) that it could be classified as an anticlockwise spiral.

6. **Merger Percentage:**
    - **Value: 0.0**

   Value: 0.0 This value further confirms the absence of recent mergers, supporting the idea that the galaxy is in a relatively stable state.

The galaxy depicted in the image is most likely a spiral galaxy with a significant stellar mass and a high likelihood of being a clockwise spiral. The data indicates that the galaxy is undisturbed and does not show signs of recent mergers or irregularities, confirming its classification as a well-formed spiral galaxy. The redshift value places it at a moderate distance from Earth.

Analysis of L2 Norm, L2 Norm with Scattering and L1 Norm with Scattering matched TNG objects

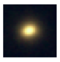
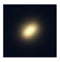
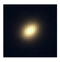
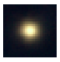
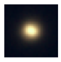
	Best 5 TNG Objects After applying L2 Norm				
Image					
TNG Data Index	1950	3208	3835	4152	6925
Redshift	0.214425035514495	0.169250233243611	0.141876203969562	0.109869940458825	0.297717684517447
Ratio Last	0.6506220987153627	0.004269939847290516	0.3315577805042267	0.5398834943771362	0.04145506769418716
DT Last	0.4842043824152156	0.1252877996906223	0.1271344419555236	0.1936592136849444	4.420015901149791
Ratio Biggest	0.6506220987153627	0.3524458408355713	0.3315577805042267	0.5398834943771362	0.04145506769418716
Snap Biggest	80	83	87	89	50
DT Biggest	0.4842043824152156	0.4966052199688376	0.1271344419555236	0.1936592136849444	4.420015901149791
Stellar Mass	0.1523748785257339	1.683871388435364	0.4315244555473328	27.86384391784668	3.066861391067505
Stellar Half Mass Radius	2.99058316421509	3.955773830413818	2.64876067962646	4.747564792633057	4.141552448272705
Stellar Mass Ratio	0.007994421757757664	0.0312962681055069	0.0163338608505085	0.2416951060295105	0.1907996982336044
New Redshift	0.0848	0.1129	0.07490000000000001	0.109869940458825	0.1184
Merger Classification	merger	merger	merger	merger	non merger

Table 4.6: L2 Norm TNG matches for SDSS object 10001374839131

The comprehensive analysis of face-on spiral galaxies uncovers a distinct and noteworthy pattern of disturbances that are effectively captured by the scattering component of both the L1 and L2 descriptors [33, 6], as referenced in 4.8 and 4.7. These disturbances manifest predominantly as asymmetric, extended features within the matched galaxies, indicating that the scattering component plays a crucial role in detecting such subtle morphological deviations. The presence of these extended features, which are likely the result of intricate gravitational interactions during merger events, underscores the importance of using scattering-based metrics in the analysis of galactic structures.

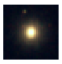
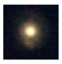
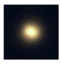
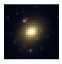
	Best 5 TNG Objects After applying L2 Norm on Scattering Transform				
Image					
TNG Data Index	624	2387	4430	6925	13282
Redshift	0.27335334657844	0.19728418237601	0.0994018026302219	0.297717684517447	0.0239744238327625
Ratio Last	0.2721417844295502	0.6438457246231649	0.6540196174512033	0.04145506769418716	0.109164923424891
DT Last	0.2356844093969119	0.4889094843587958	0.129981913608519	4.420015901149791	6.009078396746966
Ratio Biggest	0.2721417844295502	0.6438457246231649	0.6540196174512033	0.04145506769418716	0.109164923424891
Snap Biggest	78	81	90	50	60
DT Biggest	0.2356844093969119	0.4889094843587958	0.129981913608519	4.420015901149791	6.009078396746966
Stellar Mass	0.199630688619614	2.181839227676392	11.57404136657715	3.066861391067505	0.788932099342346
Stellar Half Mass Radius	2.61897745895386	3.098691940307617	9.006728172302246	4.141552448272705	2.320745706558228
Stellar Mass Ratio	0.01042638160288334	0.3370746076107025	0.1298568844795227	0.1907996982336044	0.03406895697116852
New Redshift	0.07400000000000001	0.0879	0.0994018026302219	0.1184	0.0239744238327625
Merger Classification	merger	merger	merger	non merger	non merger

Table 4.7: Scattering on L2 Norm TNG matches for SDSS object 10001374839131

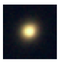
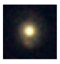
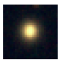
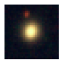
	Best 5 TNG Objects After applying L1 Norm on Scattering Transform				
Image					
TNG Data Index	295	2387	2844	3688	13005
Redshift	0.297717684517447	0.19728418237601	0.180385261705749	0.141876203969562	0.0337243718735154
Ratio Last	0.2873593868625104	0.6438457246231649	0.1141812950372696	0.9263863369465083	0.01219713035970926
DT Last	0.4614554295071982	0.4889094843587958	0.186518269664905	0.1271344419555236	2.005287368961572
Ratio Biggest	0.2873593868625104	0.6438457246231649	0.2742388248443604	0.9263863369465083	0.01328451372683048
Snap Biggest	75	81	83	87	81
DT Biggest	0.4614554295071982	0.4889094843587958	0.3713174202782152	0.1271344419555236	2.494196853320368
Stellar Mass	0.215044692158699	2.181839227676392	1.29461658000946	0.2410085946321487	16.92724770397949
Stellar Half Mass Radius	3.325106143951416	3.098691940307617	2.460989236831665	0.4662595689296722	8.757347106933594
Stellar Mass Ratio	0.009706029668450356	0.3370746076107025	0.02658739499747753	52.96695709228516	0.01931015588343143
New Redshift	0.0945	0.0879	0.0695	0.013	0.0384
Merger Classification	merger	merger	merger	merger	non merger

Table 4.8: Scattering on L1 Norm TNG matches for SDSS object 10001374839131

In contrast, the L2 norm without the inclusion of scattering, as shown in 4.6, fails to identify these anomalies, which serves to further highlight the sensitivity and effectiveness of the scattering component in detecting subtle deviations in the galaxy morphologies. This lack of detection by the standard L2 norm suggests that it may overlook important features that are critical for understanding the full extent of galactic disturbances, particularly those that arise from recent merger events. Therefore, the inclusion of scattering information becomes imperative for a more accurate and nuanced analysis of spiral galaxy morphology.

Moreover, the matching process that incorporates the scattering component consistently selects galaxies that have experienced major mergers within the last half a gigayear. This consistent selection strongly suggests a robust correlation between recent merger activity and the observed disturbances in face-on spirals. The ability of the scattering-based matching to consistently identify these recent mergers further emphasizes its effectiveness in capturing the dynamic processes that drive morphological changes in galaxies. Such findings indicate that the scattering component is not only sensitive to subtle morphological features but is also capable of linking these features to specific evolutionary events, such as recent major mergers.

The observed upward trend in DT Last values 4.4 across the various bins indicates that the matching algorithm is proficient in identifying objects with comparable merger histories. This trend implies that the algorithm is not merely capturing superficial similarities but is effectively correlating galaxies based on their deeper evolutionary trajectories. By categorizing these

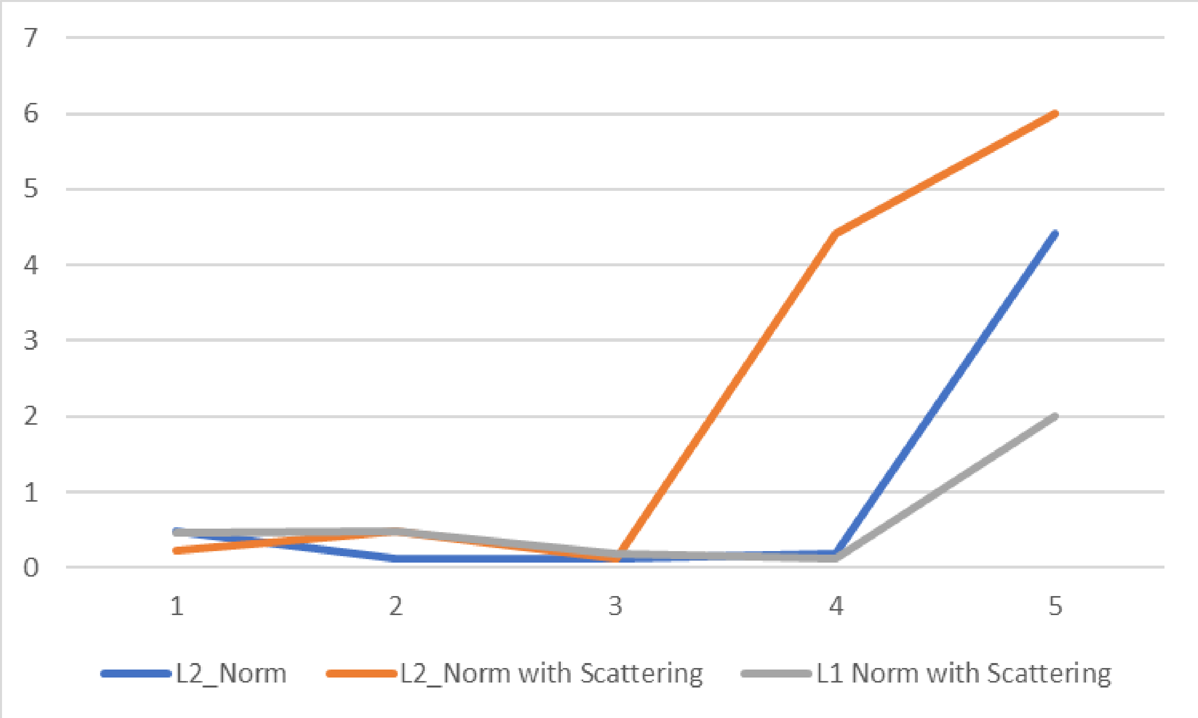


Figure 4.4: DT Last comparisons for matched TNG objects

objects into distinct bins and observing a consistent increase in DT Last values, we gain confidence in the algorithm's capacity to discern meaningful patterns in the data. This is particularly important in the context of studying galaxy evolution, as it suggests that the method is sensitive to the complex interactions and processes that shape galaxies over time.

The variations observed between the different lines—specifically, L2 Norm, L2 Norm with Scattering, and L1 Norm with Scattering—underscore the significance of incorporating scattering information and the careful selection of a distance metric when matching galaxies based on their merger histories. The L2 Norm, being a commonly used metric, provides a baseline for comparison, but the introduction of scattering and the use of the L1 Norm demonstrate that these factors can substantially alter the matching results. These differences suggest that the traditional L2 Norm may not always be sufficient for capturing the nuanced aspects of merger histories, and that alternative metrics, which take into account additional physical phenomena like scattering, can offer a more accurate representation of these histories.

4.3 Analysis of SDSS object 10000302408051

Analysis of SDSS image and data

The image 4.5 shows what appears to be a galaxy, with a bright central region and some surrounding areas of light. The data for the SDSS object 4.9 is analysed below [10]. There is also a smaller, less bright object nearby, which could be another galaxy.

1. **Stellar Mass** (Median, P16, P84, Mode, Mean): Stellar Mass Median: 10.999483 (logarithmic scale, typically in solar masses). Stellar Mass P16 (16th percentile): 10.910728. Stellar Mass P84 (84th percentile): 11.081574. Stellar Mass Mode: 11.016666. Stellar

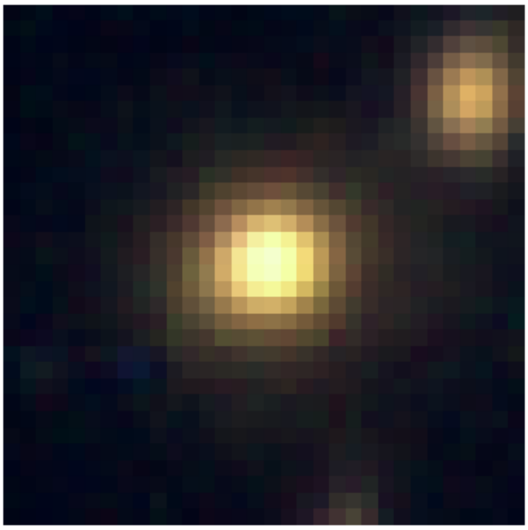


Figure 4.5: SDSS image 10000302408051

Field	Value
OBJID	10000302408051
Stellar Mass Median	10.999483
Stellar Mass P16	10.910728
Stellar Mass P84	11.081574
Stellar Mass Mode	11.016666
Stellar Mass Mean	11.009656
Redshift	0.13097674
Spectrotype	GALAXY
Subclass	AGN
Disturbed Fraction	1e+20
Disturbed Weighted Fraction	1e+20
Irregular Fraction	1e+20
Irregular Weighted Fraction	1e+20
Merger Fraction	1e+20
Merger Weighted Fraction	1e+20
Elliptical	0.57700002
Clockwise Spiral	0.0
Anticlockwise Spiral	0.037999999
Edge-on Spiral	0.115
Merger Percentage	0.231
Don't Know	0.037999999
Number of Votes	26

Table 4.9: Data for SDSS object 10000302408051

Mass Mean: 11.009656. These values indicate the distribution of the galaxy's stellar mass, likely in logarithmic units of solar masses. The median, mean, mode, and percentiles give a statistical summary of the stellar mass distribution.

2. **Redshift:** 0.13097674 The redshift value indicates how much the light from the galaxy has been stretched due to the expansion of the universe. This value can be used to estimate the distance of the galaxy from Earth.
3. **Spectrotype**: **GALAXY**: The object is classified as a galaxy.
4. **Subclass**: **AGN (Active Galactic Nucleus)**: This indicates that the galaxy hosts an active galactic nucleus, a region at its center with a supermassive black hole that is actively accreting material and emitting large amounts of energy.
5. **Disturbed, Irregular, and Merger Fractions**: The values for **Disturbed Fraction**, **Irregular Fraction**, and **Merger Fraction** are all 1e+20, which seems unusually high, suggesting there might be an issue with the data or that these fields were not accurately measured.
6. **Elliptical:** 0.57700002 This value represents the likelihood that the galaxy has an elliptical shape.
7. **Clockwise Spiral:** 0.0 Indicates there is no evidence of the galaxy being a clockwise spiral.
8. **Anticlockwise Spiral:** 0.037999999 Indicates a very low probability of the galaxy being an anticlockwise spiral.
9. **Edge-on Spiral:** 0.115 There is a 11.5% chance that the galaxy is viewed edge-on, suggesting a disk-like structure might be present.
10. **Merger Percentage:** 0.231 This value indicates a 23.1% likelihood that this galaxy is involved in a merger with another galaxy.
11. **Don't Know:** 0.037999999 A small percentage of responses (3.8%) were uncertain about the classification of this galaxy.
12. **Number of Votes:** 26 This indicates that 26 votes or classifications were made regarding this galaxy's features, likely from a citizen science project like Galaxy Zoo.

The image and data suggest this is a galaxy with an active galactic nucleus (AGN) and a significant likelihood of being elliptical. There is also some indication of a merger or interaction with another galaxy. The redshift places it at a moderate distance from Earth, and the data includes various probabilities about its morphology based on classifications.

Analysis of L2 Norm, L2 Norm with Scattering and L1 Norm with Scattering matched TNG objects

In the case of this particular SDSS object [10], the two scattering models, L1 and L2 [6, 33], exhibit distinct behaviors 4.10 ,4.11. The comparison between these two models reveals significant differences in their approach to identifying and matching galaxy features, each with its strengths and limitations.

The L2 scattering model demonstrates a heightened sensitivity to slightly red features, which are relatively uncommon in the dataset. This is evident from the consistently higher "Ratio Last" values in Table 4.10 compared to those in Table 4.11 for the L1 scattering. However, this sensitivity comes at a cost. The L2 scattering model shows a tendency to match visually dissimilar galaxies, as seen in examples like match 2 in Table 4.10. This inconsistency suggests that while the L2 scattering model may excel at identifying rare color-based features, it struggles to capture the overall structural characteristics of galaxies consistently. In contrast, the L1

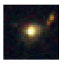
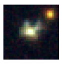
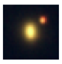
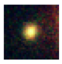
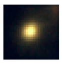
	Best 5 TNG Objects After applying L2 Norm on Scattering Transform				
Image					
TNG Data Index	28	1526	4803	8381	11712
Redshift	0.297717684517447	0.225988386260198	0.0838844307974793	0.225988386260198	0.0838844307974793
Ratio Last	0.6530154347419739	0.6886216402053833	0.6369836330413818	0.01947482861578465	0.03223235532641411
DT Last	0.4614554295071982	0.1818784280492518	0.1962649977766309	5.999275616814802	2.825432910389353
Ratio Biggest	0.6530154347419739	0.6886216402053833	0.6369836330413818	0.01947482861578465	0.03223235532641411
Snap Biggest	75	81	91	45	75
DT Biggest	0.4614554295071982	0.1818784280492518	0.1962649977766309	5.999275616814802	2.825432910389353
Stellar Mass	0.1613916009664536	0.2666714787483215	0.1927099525928497	0.3999189734458923	1.264570951461792
Stellar Half Mass Radius	0.3713125884532928	0.7265868782997131	0.4449168145656586	2.219445943832397	2.502562761306763
Stellar Mass Ratio	106.4081039428711	75.35178375244414	63.52840042114258	0.2586672056823059	0.04167857766151428
New Redshift	0.0103	0.0202	0.0124	0.0626	0.07070000000000001
Merger Classification	merger	merger	merger	non merger	non merger

Table 4.10: Scattering on L2 Norm TNG matches for SDSS object 10000302408051

scattering model exhibits superior performance in capturing structural attributes of galaxies. The matched galaxies in Table 4.11 appear more visually similar, indicating better structural alignment. This suggests that the L1 scattering is less affected by minor colour variations and instead focuses more on the overall galaxy morphology. This robustness in structural matching implies that the L1 scattering model may be more reliable for automated galaxy classification tasks, especially when morphological features are the primary concern.

A closer examination of the quantitative data reveals further insights. The L1 scattering matches show more consistent Stellar Mass values compared to the L2 scattering matches. Similarly, the DT Biggest parameter exhibits more stable values across matches in the L1 scattering model. These observations support the notion that the L1 scattering provides a more reliable representation of galaxy structures in the SDSS dataset.

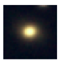
	Best 5 TNG Objects After applying L1 Norm on Scattering Transform				
Image					
TNG Data Index	1665	2662	4064	6116	12649
Redshift	0.225988386260198	0.180385261705749	0.125759332411261	0.0239744283827625	0.0485236299818059
Ratio Last	0.3731016516685486	0.03409386798739433	0.8914813793518578	0.5975404381752014	0.02513047680258751
DT Last	0.3619724767189818	0.3713174202782152	0.3191918268944828	0.3361575745796345	3.901636313154246
Ratio Biggest	0.3731016516685486	0.5207038004773809	0.8914813793518578	0.5975404381752014	0.02513047680258751
Snap Biggest	80	82	87	95	71
DT Biggest	0.3619724767189818	0.4935493259744449	0.3191918268944828	0.3361575745796345	3.901636313154246
Stellar Mass	0.7949736502502441	20.35225677490234	0.4893104434013367	0.1522184163331985	0.5219466966481018
Stellar Half Mass Radius	2.779541254043579	11.64157676696777	3.208225965499878	0.2735964357852936	1.770488381385803
Stellar Mass Ratio	0.02073444612324238	0.02721228285656767	0.01095079164952204	100.3600769042969	0.07883805036544448
New Redshift	0.0787	0.180385261705749	0.0911	0.007600000000000001	0.0485236299818059
Merger Classification	merger	merger	merger	merger	non merger

Table 4.11: Scattering on L1 Norm TNG matches for SDSS object 10000302408051

Interestingly, both models consistently classify the matches as "merger" candidates. This agreement suggests that the merger classification is robust across different scattering norms, which could be valuable for studying galaxy evolution and interactions. However, the models differ in their redshift estimations, with L1 scattering matches showing higher redshift values compared to L2 scattering matches. This difference could have significant implications for the interpretation of galaxy properties and evolution across cosmic time.

The findings from this analysis highlight the crucial role of scattering norm selection in galaxy classification and morphological studies. While the L2 [33] scattering [6, 3] shows promise in identifying rare colour features, the L1 [33] scattering[6, 3] demonstrates superior performance in capturing and consistently matching structural attributes of galaxies. This insight can guide researchers in choosing the most appropriate model for their specific astrophysical investigations.

Chapter 5

Conclusions and Future work

5.1 Summary of Challenges in Galaxy Merger Classification

The complexities of galaxy merger classification have long been a subject of interest for astronomers, as galaxy mergers play a pivotal role in understanding the evolution and behavior of galaxies [8, 7]. Chapter 1 outlined the various issues plaguing accurate and consistent classification of galaxy mergers, including the irregular and distorted morphologies that emerge during different phases of the merging process [12]. These morphologies make it difficult to classify galaxies using traditional methods, especially when their forms deviate substantially from typical galactic structures. As we have seen, the primary factors contributing to this challenge include the different stages of mergers, which present unique and varying observable features that often complicate classification efforts [8, 1]. The diversity of mass ratios between merging galaxies and projection effects from different observational angles further obfuscate the task. Environmental interactions and redshift effects also play a role in creating structures that mimic merger signatures, adding even more layers of complexity to an already intricate process. Furthermore, the subjective nature of visual classifications exacerbates these difficulties, leading to inconsistent categorization outcomes between different observers [11].

These challenges indicate the need for more advanced techniques beyond mere visual inspection, which can be prone to human error and limitations in perspective. With this in mind, it is essential to leverage technological advancements, particularly in the realm of computational and analytical tools, to enhance the precision of galaxy merger classification.

5.2 The Role of Simulations in Addressing Real Data Limitations

One of the key insights gained from this project is the importance of simulated data in overcoming the limitations inherent in real galaxy observations [10, 8]. Unlike real galaxies, which can only be observed from a single perspective with no historical data to reference, simulated galaxies offer a wealth of information [9, 5]. As discussed in Chapter 1, simulations enable us to fully explore the past, present, and future interactions of galaxies, providing a complete and detailed history of their mergers. This temporal flexibility allows researchers to predict future mergers with greater accuracy and analyze long-term galactic evolution processes that would otherwise take billions of years to observe in real time.

Moreover, simulations allow us to generate images of galaxies from multiple angles, offering an unparalleled level of detail and flexibility for morphological studies [25]. By viewing galaxies from various perspectives, astronomers can identify structural features and potential interactions that would be missed in a single-angle observation. This multidimensional analysis proves invaluable for galaxy classification, as it allows researchers to compare simulated mergers with real galaxies and better understand their interactions. By bridging the gap between real and simulated data, scientists are able to refine their methods for detecting and analyzing galaxy mergers, ultimately improving classification accuracy.

5.3 The Utility of Scattering Transform Networks

A major technological novelty discussed in this work is the application of scattering transform networks [3, 6] to the classification of galaxy mergers. Scattering transforms offer distinct advantages over traditional methods, especially in cases where real galaxy data is limited in quality or resolution. As noted in Chapter 1, the translation invariance of scattering transforms makes them robust against varying observation angles, an essential feature when dealing with galactic images taken from different perspectives [15]. These transforms excel in multiscale analysis, capturing features across various spatial scales and enhancing discrimination between different stages of mergers.

Scattering transform networks also facilitate dimensionality reduction, maintaining robustness to noise in observational data while producing interpretable intermediate representations that offer insights into the physical processes underlying galaxy mergers [13]. The method is particularly well-suited to low-resolution images, which are common in real-world datasets such as those sourced from the Sloan Digital Sky Survey (SDSS) [10]. By focusing on the edges of galaxies and using distance calculations between edges, scattering transforms avoid the pitfalls that often affect traditional convolutional neural networks (CNNs) or transformer models, which require higher-quality images to function optimally [12].

Despite its promise, scattering transform networks are not without challenges, as demonstrated by the results obtained. Background edges in images can mislead the algorithm, leading to incorrect classifications. However, with proper dataset preparation, such as background sky subtraction [1] in the case of galaxy images, scattering transforms can be a powerful tool for comparing real and simulated images of mergers.

After observing the results it becomes clear that Scattering transforms although capable of matching objects that are similar still pose an issue for matching astronomical objects. It is observed that the matched TNG galaxies [9] have a similar 2-dimensional orientation when compared to the original SDSS image [10]. Yet two galaxies which have similar properties may not have similar 2D orientation. This introduces the issue of missing out on matches which might have similar edges but have a different 2-dimensional orientation.

5.4 Future Directions for Galaxy Merger Research

The insights gained from this project point toward several promising directions for future research. First, the continued refinement of scattering transform networks will be crucial for improving the accuracy and reliability of galaxy merger classification. Future work should focus on fine-tuning these networks to better handle background noise and complex galaxy morphologies.

The Scattering Transform issue of ignoring matches with different 2D orientations can be solved by introducing the ability to maintain invariance of the galaxy orientaion while passing it through the Scattering transform [15]. This may be achieved by rotating the image once filtered in multiple degrees and summing the results together.

Moreover, the integration of machine learning models with scattering transforms could enhance the overall classification framework. Hybrid models that combine the strengths of CNNs, scattering transforms, and other deep learning techniques could further improve performance, especially in cases where large datasets of real galaxies are available [6, 13, 1]. In particular, leveraging unsupervised learning methods could help to identify new features or patterns in galaxy mergers that are not readily apparent with current techniques [12].

5.5 Implications for Broader Astronomical Studies

The advancements in galaxy merger classification have broad implications for the field of astronomy. Accurately identifying mergers is critical for understanding galaxy evolution, and the tools and techniques discussed in this project will play a significant role in advancing our knowledge of how galaxies grow and change over time [8, 12, 7]. In particular, improving the classification of galaxy mergers will enable astronomers to better understand the role of mergers in triggering star formation, the formation of supermassive black holes, and the overall structure of the universe.

Furthermore, the ability to more accurately classify galaxy mergers has potential implications for other areas of astronomy, including the study of dark matter and dark energy. Galaxy mergers provide a natural laboratory for testing theories of dark matter, as the dynamics of merging galaxies can offer insights into the distribution and behaviour of dark matter in the universe. Similarly, understanding how mergers contribute to the growth and evolution of galaxies may shed light on the influence of dark energy in driving cosmic expansion.

5.6 Concluding Remarks

In conclusion, the classification of galaxy mergers remains a formidable challenge, but the development of advanced analytical techniques such as scattering transform networks offers new hope for overcoming these difficulties. By leveraging the strengths of simulations and computational models, astronomers can improve the precision and reliability of galaxy merger classification, leading to deeper insights into the dynamic and multifaceted nature of galaxy evolution. While challenges remain, particularly in dealing with low-resolution images and complex morphologies, the future of galaxy merger research is bright. Continued advancements in both simulation and machine learning technologies will undoubtedly push the field forward, providing the tools necessary to unlock the mysteries of the universe and the forces that shape it.

Number of words until this point, excluding front matter: 10000.

Bibliography

[1] M. S. Avirett-Mackenzie, C. Villforth, M. Huertas-Company, S. Wuyts, D. M. Alexander, S. Bonoli, A. Lapi, I. E. Lopez, C. Ramos Almeida, and F. Shankar, "A post-merger enhancement only in star-forming Type 2 Seyfert galaxies: the deep learning view," *Monthly Notices of the Royal Astronomical Society*, vol. 528, no. 4, pp. 6915–6933, Mar. 2024. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2024MNRAS.528.6915A

[2] University of Manchester. (2023) Astronomers discover how galaxies form through mergers. Accessed: 2024-09-05. [Online]. Available: https://www.manchester.ac.uk/about/news/astronomers-discover-how-galaxies-form-through-mergers/

[3] S. Mallat, "Group invariant scattering," 2012. [Online]. Available: https://arxiv.org/abs/1101.2286

[4] V. Springel, "E pur si muove: Galilean-invariant cosmological hydrodynamical simulations on a moving mesh," *Monthly Notices of the Royal Astronomical Society*, vol. 401, no. 2, pp. 791–851, 01 2010. [Online]. Available: https://doi.org/10.1111/j.1365-2966.2009.15715.x

[5] P. F. Hopkins, L. Hernquist, T. J. Cox, T. D. Matteo, B. Robertson, and V. Springel, "A unified, merger-driven model of the origin of starbursts, quasars, the cosmic x-ray background, supermassive black holes, and galaxy spheroids," *The Astrophysical Journal Supplement Series*, vol. 163, no. 1, p. 1, mar 2006. [Online]. Available: https://dx.doi.org/10.1086/499298

[6] M. Eickenberg, G. Exarchakis, M. Hirn, S. Mallat, and L. Thiry, "Solid harmonic wavelet scattering for predictions of molecule properties," *The Journal of Chemical Physics*, vol. 148, no. 24, p. 241732, 05 2018. [Online]. Available: https://doi.org/10.1063/1.5023798

[7] C. Bottrell, M. H. Hani, H. Teimoorinia, S. L. Ellison, J. Moreno, P. Torrey, C. C. Hayward, M. Thorp, L. Simard, and L. Hernquist, "Deep learning predictions of galaxy merger stage and the importance of observational realism," *Monthly Notices of the Royal Astronomical Society*, vol. 490, no. 4, p. 5390–5413, Oct. 2019. [Online]. Available: http://dx.doi.org/10.1093/mnras/stz2934

[8] C. J. Conselice, "Galaxy formation: Where do we stand?" 2012. [Online]. Available: https://arxiv.org/abs/1212.5641

[9] D. Nelson, V. Springel, A. Pillepich, V. Rodriguez-Gomez, P. Torrey, S. Genel, M. Vogelsberger, R. Pakmor, F. Marinacci, R. Weinberger, L. Kelley, M. Lovell, B. Diemer, and L. Hernquist, "The IllustrisTNG simulations: public data release," *Computational Astrophysics and Cosmology*, vol. 6, no. 1, p. 2, May 2019. [Online]. Available: https://doi.org/10.1186/s40668-019-0028-x

[10] A. Almeida, S. F. Anderson, M. Argudo-Fernández, C. Badenes, K. Barger, J. K. Barrera-Ballesteros, C. F. Bender, E. Benitez, F. Besser, J. C. Bird, D. Bizyaev, M. R. Blanton, J. Bochanski, J. Bovy, W. N. Brandt, J. R. Brownstein, J. Buchner, E. Bulbul, J. N. Burchett, M. C. Díaz, J. K. Carlberg, A. R. Casey, V. Chandra, B. Cherinka, C. Chiappini, A. A. Coker, J. Comparat, C. Conroy, G. Contardo, A. Cortes, K. Covey, J. D. Crane, K. Cunha, C. Dabbieri, J. W. Davidson, M. C. Davis, A. B. de Andrade Queiroz, N. De Lee, J. E. Méndez Delgado, S. Demasi, F. Di Mille, J. Donor, P. Dow, T. Dwelly, M. Eracleous, J. Eriksen, X. Fan, E. Farr, S. Frederick, L. Fries, P. Frinchaboy, B. T. Gänsicke, J. Ge, C. González Ávila, K. Grabowski, C. Grier, G. Guiglion, P. Gupta, P. Hall, K. Hawkins, C. R. Hayes, J. J. Hermes, L. Hernández-García, D. W. Hogg, J. A. Holtzman, H. J. Ibarra-Medel, A. Ji, P. Jofre, J. A. Johnson, A. M. Jones, K. Kinemuchi, M. Kluge, A. Koekemoer, J. A. Kollmeier, M. Kounkel, D. Krishnarao, M. Krumpe, I. Lacerna, P. J. A. Lago, C. Laporte, C. Liu, A. Liu, X. Liu, A. R. Lopes, M. Macktoobian, S. R. Majewski, V. Malanushenko, D. Maoz, T. Masseron, K. L. Masters, G. Matijevic, A. McBride, I. Medan, A. Merloni, S. Morrison, N. Myers, S. Mészáros, C. A. Negrete, D. L. Nidever, C. Nitschelm, D. Oravetz, A. Oravetz, K. Pan, Y. Peng, M. H. Pinsonneault, R. Pogge, D. Qiu, S. V. Ramirez, H.-W. Rix, D. F. Rosso, J. Runnoe, M. Salvato, S. F. Sanchez, F. A. Santana, A. Saydjari, C. Sayres, K. C. Schlaufman, D. P. Schneider, A. Schwope, J. Serna, Y. Shen, J. Sobeck, Y.-Y. Song, D. Souto, T. Spoo, K. G. Stassun, M. Steinmetz, I. Straumit, G. Stringfellow, J. Sánchez-Gallego, M. Taghizadeh-Popp, J. Tayar, A. Thakar, P. B. Tissera, A. Tkachenko, H. H. Toledo, B. Trakhtenbrot, J. G. Fernández-Trincado, N. Troup, J. R. Trump, S. Tuttle, N. Ulloa, J. A. Vazquez-Mata, P. V. Alfaro, S. Villanova, S. Wachter, A.-M. Weijmans, A. Wheeler, J. Wilson, L. Wojno, J. Wolf, X.-X. Xue, J. E. Ybarra, E. Zari, and G. Zasowski, "The eighteenth data release of the sloan digital sky surveys: Targeting and first spectra from sdss-v," *The Astrophysical Journal Supplement Series*, vol. 267, no. 2, p. 44, Aug. 2023. [Online]. Available: http://dx.doi.org/10.3847/1538-4365/acda98

[11] S. Koppula, V. Bapst, M. Huertas-Company, S. Blackwell, A. Grabska-Barwinska, S. Dieleman, A. Huber, N. Antropova, M. Binkowski, H. Openshaw, A. Recasens, F. Caro, A. Deke, Y. Dubois, J. V. Ferrero, D. C. Koo, J. R. Primack, and T. Back, "A deep learning approach for characterizing major galaxy mergers," 2021. [Online]. Available: https://arxiv.org/abs/2102.05182

[12] R. Nevin, L. Blecha, J. Comerford, and J. Greene, "Accurate identification of galaxy mergers with imaging," *The Astrophysical Journal*, vol. 872, no. 1, p. 76, feb 2019. [Online]. Available: https://dx.doi.org/10.3847/1538-4357/aafd34

[13] Delouis, J.-M., Allys, E., Gauvrit, E., and Boulanger, F., "Non-gaussian modelling and statistical denoising of planck dust polarisation full-sky maps using scattering transforms," *A&A*, vol. 668, p. A122, 2022. [Online]. Available: https://doi.org/10.1051/0004-6361/202244566

[14] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009. [Online]. Available: https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[15] A. K. Saydjari and D. P. Finkbeiner, "Equivariant wavelets: Fast rotation

and translation invariant wavelet scattering transforms," 2021. [Online]. Available: https://arxiv.org/abs/2104.11244

[16] V. Morgenshtern, "Lecture 21: Scattering transform," 2019. [Online]. Available: https://www.machinelearning.tf.fau.de/pdfs/mmmlisp19/7-scattering.pdf

[17] V. Springel, R. Pakmor, A. Pillepich, R. Weinberger, D. Nelson, L. Hernquist, M. Vogelsberger, S. Genel, P. Torrey, F. Marinacci, and J. Naiman, "First results from the illustristng simulations: matter and galaxy clustering," *Monthly Notices of the Royal Astronomical Society*, vol. 475, no. 1, p. 676–698, Dec. 2017. [Online]. Available: http://dx.doi.org/10.1093/mnras/stx3304

[18] A. Pillepich, D. Nelson, L. Hernquist, V. Springel, R. Pakmor, P. Torrey, R. Weinberger, S. Genel, J. P. Naiman, F. Marinacci, and M. Vogelsberger, "First results from the illustristng simulations: the stellar mass content of groups and clusters of galaxies," *Monthly Notices of the Royal Astronomical Society*, vol. 475, no. 1, p. 648–675, Dec. 2017. [Online]. Available: http://dx.doi.org/10.1093/mnras/stx3112

[19] J. P. Naiman, A. Pillepich, V. Springel, E. Ramirez-Ruiz, P. Torrey, M. Vogelsberger, R. Pakmor, D. Nelson, F. Marinacci, L. Hernquist, R. Weinberger, and S. Genel, "First results from the illustristng simulations: a tale of two elements – chemical evolution of magnesium and europium," *Monthly Notices of the Royal Astronomical Society*, vol. 477, no. 1, p. 1206–1224, Mar. 2018. [Online]. Available: http://dx.doi.org/10.1093/mnras/sty618

[20] D. Nelson, A. Pillepich, V. Springel, R. Weinberger, L. Hernquist, R. Pakmor, S. Genel, P. Torrey, M. Vogelsberger, G. Kauffmann, F. Marinacci, and J. Naiman, "First results from the illustristng simulations: the galaxy colour bimodality," *Monthly Notices of the Royal Astronomical Society*, vol. 475, no. 1, p. 624–647, Nov. 2017. [Online]. Available: http://dx.doi.org/10.1093/mnras/stx3040

[21] F. Marinacci, M. Vogelsberger, R. Pakmor, P. Torrey, V. Springel, L. Hernquist, D. Nelson, R. Weinberger, A. Pillepich, J. Naiman, and S. Genel, "First results from the illustristng simulations: radio haloes and magnetic fields," *Monthly Notices of the Royal Astronomical Society*, Aug. 2018. [Online]. Available: http://dx.doi.org/10.1093/mnras/sty2206

[22] A. Pillepich, D. Nelson, V. Springel, R. Pakmor, P. Torrey, R. Weinberger, M. Vogelsberger, F. Marinacci, S. Genel, A. van der Wel, and L. Hernquist, "First results from the tng50 simulation: the evolution of stellar and gaseous discs across cosmic time," *Monthly Notices of the Royal Astronomical Society*, vol. 490, no. 3, p. 3196–3233, Sep. 2019. [Online]. Available: http://dx.doi.org/10.1093/mnras/stz2338

[23] D. Nelson, A. Pillepich, V. Springel, R. Pakmor, R. Weinberger, S. Genel, P. Torrey, M. Vogelsberger, F. Marinacci, and L. Hernquist, "First results from the tng50 simulation: galactic outflows driven by supernovae and black hole feedback," *Monthly Notices of the Royal Astronomical Society*, vol. 490, no. 3, p. 3234–3261, Aug. 2019. [Online]. Available: http://dx.doi.org/10.1093/mnras/stz2306

[24] R. Weinberger, V. Springel, L. Hernquist, A. Pillepich, F. Marinacci, R. Pakmor, D. Nelson, S. Genel, M. Vogelsberger, J. Naiman, and P. Torrey, "Simulating galaxy formation with black hole driven thermal and kinetic feedback," *Monthly Notices of*

the Royal Astronomical Society, vol. 465, no. 3, p. 3291–3308, Nov. 2016. [Online]. Available: http://dx.doi.org/10.1093/mnras/stw2944

[25] A. Pillepich, V. Springel, D. Nelson, S. Genel, J. Naiman, R. Pakmor, L. Hernquist, P. Torrey, M. Vogelsberger, R. Weinberger, and F. Marinacci, "Simulating galaxy formation with the illustristng model," *Monthly Notices of the Royal Astronomical Society*, vol. 473, no. 3, p. 4077–4106, Oct. 2017. [Online]. Available: http://dx.doi.org/10.1093/mnras/stx2656

[26] T. A. Collaboration, A. M. Price-Whelan, P. L. Lim, N. Earl, N. Starkman, L. Bradley, D. L. Shupe, A. A. Patil, L. Corrales, C. E. Brasseur, M. Nöthe, A. Donath, E. Tollerud, B. M. Morris, A. Ginsburg, E. Vaher, B. A. Weaver, J. Tocknell, W. Jamieson, M. H. van Kerkwijk, T. P. Robitaille, B. Merry, M. Bachetti, H. M. Günther, T. L. Aldcroft, J. A. Alvarado-Montes, A. M. Archibald, A. Bódi, S. Bapat, G. Barentsen, J. Bazán, M. Biswas, M. Boquien, D. J. Burke, D. Cara, M. Cara, K. E. Conroy, S. Conseil, M. W. Craig, R. M. Cross, K. L. Cruz, F. D'Eugenio, N. Dencheva, H. A. R. Devillepoix, J. P. Dietrich, A. D. Eigenbrot, T. Erben, L. Ferreira, D. Foreman-Mackey, R. Fox, N. Freij, S. Garg, R. Geda, L. Glattly, Y. Gondhalekar, K. D. Gordon, D. Grant, P. Greenfield, A. M. Groener, S. Guest, S. Gurovich, R. Handberg, A. Hart, Z. Hatfield-Dodds, D. Homeier, G. Hosseinzadeh, T. Jenness, C. K. Jones, P. Joseph, J. B. Kalmbach, E. Karamehmetoglu, M. Kałuszyński, M. S. P. Kelley, N. Kern, W. E. Kerzendorf, E. W. Koch, S. Kulumani, A. Lee, C. Ly, Z. Ma, C. MacBride, J. M. Maljaars, D. Muna, N. A. Murphy, H. Norman, R. O'Steen, K. A. Oman, C. Pacifici, S. Pascual, J. Pascual-Granado, R. R. Patil, G. I. Perren, T. E. Pickering, T. Rastogi, B. R. Roulston, D. F. Ryan, E. S. Rykoff, J. Sabater, P. Sakurikar, J. Salgado, A. Sanghi, N. Saunders, V. Savchenko, L. Schwardt, M. Seifert-Eckert, A. Y. Shih, A. S. Jain, G. Shukla, J. Sick, C. Simpson, S. Singanamalla, L. P. Singer, J. Singhal, M. Sinha, B. M. Sipőcz, L. R. Spitler, D. Stansby, O. Streicher, J. Šumak, J. D. Swinbank, D. S. Taranu, N. Tewary, G. R. Tremblay, M. d. Val-Borro, S. J. Van Kooten, Z. Vasović, S. Verma, J. V. de Miranda Cardoso, P. K. G. Williams, T. J. Wilson, B. Winkel, W. M. Wood-Vasey, R. Xue, P. Yoachim, C. Zhang, and A. Zonca, "The astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package*," *The Astrophysical Journal*, vol. 935, no. 2, p. 167, Aug. 2022. [Online]. Available: http://dx.doi.org/10.3847/1538-4357/ac7c74

[27] The Astropy Collaboration, Robitaille, Thomas P., Tollerud, Erik J., Greenfield, Perry, Droettboom, Michael, Bray, Erik, Aldcroft, Tom, Davis, Matt, Ginsburg, Adam, Price-Whelan, Adrian M., Kerzendorf, Wolfgang E., Conley, Alexander, Crighton, Neil, Barbary, Kyle, Muna, Demitri, Ferguson, Henry, Grollier, Frédéric, Parikh, Madhura M., Nair, Prasanth H., Günther, Hans M., Deil, Christoph, Woillez, Julien, Conseil, Simon, Kramer, Roban, Turner, James E. H., Singer, Leo, Fox, Ryan, Weaver, Benjamin A., Zabalza, Victor, Edwards, Zachary I., Azalee Bostroem, K., Burke, D. J., Casey, Andrew R., Crawford, Steven M., Dencheva, Nadia, Ely, Justin, Jenness, Tim, Labrie, Kathleen, Lim, Pey Lian, Pierfederici, Francesco, Pontzen, Andrew, Ptak, Andy, Refsdal, Brian, Servillat, Mathieu, and Streicher, Ole, "Astropy: A community python package for astronomy," *A&A*, vol. 558, p. A33, 2013. [Online]. Available: https://doi.org/10.1051/0004-6361/201322068

[28] Pence, W. D., Chiappetti, L., Page, C. G., Shaw, R. A., and Stobie, E., "Definition of

the flexible image transport system (fits), version 3.0," *A&A*, vol. 524, p. A42, 2010. [Online]. Available: https://doi.org/10.1051/0004-6361/201015362

[29] F. Ochsenbein, "Astronomical catalogues and tables adopted standards," 2000.

[30] I. M. George and L. Angelini, "Specification of physical units within ogip fits les," OGIP Memo OGIP/93-001, NASA Goddard Space Flight Center, Greenbelt, Maryland ..., Tech. Rep., 1995.

[31] M. Demleitner, S. Derrière, N. Gray, M. Louys, and F. Ochsenbein, "Units in the vo, version 1.0," *IVOA Recommendation. http://www. ivoa. net/documents/VOUnits/index. html*, 2014.

[32] M. Andreux, T. Angles, G. Exarchakis, R. Leonarduzzi, G. Rochette, L. Thiry, J. Zarka, S. Mallat, J. andén, E. Belilovsky, J. Bruna, V. Lostanlen, M. Chaudhary, M. J. Hirn, E. Oyallon, S. Zhang, C. Cella, and M. Eickenberg, "Kymatio: Scattering transforms in python," 2022. [Online]. Available: https://arxiv.org/abs/1812.11214

[33] S. Bektaş and Y. Şişman, "The comparison of l1 and l2-norm minimization methods," *International Journal of the Physical Sciences*, vol. 5, no. 11, pp. 1721–1727, 2010.

Appendix A

Code

A.1 Github Repository for Jupyter notebooks for implementation of scattering transform

https://github.com/rwitobaansheikh/Scattering_on_galaxies

A.2 Github Repository for code of Flask webapp

https://github.com/rwitobaansheikh/Scattering_on_galaxies_Webapp

Appendix B

Web app link

https://scattering-on-galaxies-d2f545dbfe50.herokuapp.com/